Measuring Normative Information

What a value-alignment dataset can and cannot tell you, derived from zero

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Abstract

A public dataset of human value judgments ships 15,248 written criteria with numerical weights, and 82.3% of those criteria contain no condition, no exception, no priority, no hedge and no prohibition – the collected normative payload is content, a sign, and a magnitude. This paper asks what can be measured about such an object at all, and answers it as a typed question: every quantity is defined relative to a declared query family, a declared decision family, and a named instrument, and each is derived here from nothing, assuming no statistics. What survives is: (i) a theorem that no finite weight encodes a veto, because the penalty reproducing one depends on which candidate responses happened to be shown, with the counterexample present in the data at 2.65× the elicitation scale; (ii) that a person's own written criteria predict a *stranger's* ranking at 0.310 and their own at 0.393, so the individual signal in the elicitation is 0.083 of cosine and nothing downstream can destroy what was never collected; (iii) that comparative measurements are invariant across three executors and a presentation-order swap (Spearman 0.963–1.000) while absolute ones are not (40% of winners change with the reader, 8 points with the page order); and (iv) that one bit per criterion – the sign alone – reproduces 89.8% of the shipped decisions. Six claims were withdrawn during this work and are listed with what killed them; five of the six died the same way, to a verdict decided by a constant the author chose rather than by a comparison against a range of defensible references. The central claim ends if a person's induced ordering proves unstable across response sets: what was elicited would then be a property of the probe rather than of the person, and no measurement downstream would be about values at all.

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1 Introduction

1.1 The question, and why the obvious version of it is the wrong one

The obvious question is: *how much of what people said survives the pipeline?* It has no answer, and the reason is not empirical.

To say that two normative objects carry "the same information" one must fix what one intends to ask of them. Proposition 111 shows that the choice of question family determines the answer completely: under one admissible family every change is total loss, under another no change is any loss. A framework that reports a single percentage has therefore not measured a property of the pipeline; it has reported a choice it did not disclose.

The question this paper answers instead is: *given a declared family of decisions, what is identified, what is bounded, and what is forbidden?* That version has answers, several of them are theorems rather than measurements, and the largest of them turns out to concern the elicitation rather than the pipeline.

1.2 Scope, identity, and contributions

The object is a released dataset of human comparative judgments over four candidate responses per prompt, together with written criteria and numeric weights, and a compiled "core" standard derived from them (§10). The satisfaction of each criterion by each response is not in the release; it is reconstructed with a language-model judge, and every consequence of that is tracked explicitly (§15).

Contributions, in the order they appear:

1. A self-contained derivation of every statistical tool used, from the definition of probability through the design effect, attenuation and minimum detectable effect (§8). No external statistical background is assumed at any point.
2. The probe bound (theorem 74): the elicitation returns two free parameters per person per prompt, whatever the criteria say.
3. A measurement that the weight scale is used as an *interval* scale, with the fraction of decisions that depend on it (§11.2).
4. The social-choice structure of the corpus, including the finding that the weighted-criteria route agrees with the Condorcet winner only 68.4% of the time while every ranking rule agrees 97–98% (§12).
5. theorem 90 and theorem 91: the exact condition under which a penalty reproduces a veto, and the proof that no finite penalty does so uniformly.
6. Normalised regret (definition 113) as the decision-theoretically correct estimand, with the exact identity showing the flip rate to be its coarsening (lemma 115).
7. The gauge result separating comparative from absolute claims (§15.3).
8. An explicit non-identification section with proofs (§22), and a design specification derived line by line from it (§28).

1.3 Relation to neighbouring work

Everything in §8 is standard and is reproduced only so the paper is self-contained. Blackwell's comparison of experiments, Arrow's impossibility theorem, Sen's liberal paradox, the Shapley value, the Spearman–Brown prophecy formula, the design effect and classical attenuation are all classical; they are derived rather than cited because a reader with no statistical background cannot otherwise check the argument. The novelty boundary is stated explicitly in §24 and the Discussion: what is new is the application to this object and five specific measured quantities, not the machinery.

1.4 Roadmap

§8 builds the mathematics. §10 defines the object exactly. §18.1 states the measurement contract. §11–§15 walk the chain one arrow at a time. §20 and §21 give the estimators. §22 proves what cannot be identified. §23 states the findings with their confidence cards, §24 what did not survive, §25 what would end each claim, and §28 what a sufficient design requires.

2 What normative information is

why The phrase is used in every alignment paper and defined in none. Six definitions are in circulation, they disagree by orders of magnitude on the same data, and five are refutable by construction. This section fixes the object before anything is measured, because proposition 111 proves the measurement is determined by the choice.

2.1 The naive picture, and the exact point of failure

the picture	values → writing → aggregate → standard → model; a fraction is lost at each arrow and can be reported
the failure	"fraction *of what*" has no referent. Values are a relation on an infinite set, not a substance with a quantity
what is needed	a criterion for two normative objects being *the same*, which is a modelling decision with at least six defensible answers

2.2 Six candidate definitions, five refuted by construction

	definition	refuting construction	measured here
C1	the *text*; loss = words that did not survive	paraphrase is total loss, verbatim copy into an inapplicable context is zero loss	compilation rewrites every criterion, so C1 scores 100% loss for an object that still agrees on **70.6%** of decisions
C2	the *numbers*; loss = weights that changed	multiply every weight by 2: all decisions, rankings and aggregates identical, C2 says everything changed	the scale has no shared zero, no shared unit, and an interval reading worth **11.7%** of decisions (§11.2)
C3	the *induced ranking*	an ordering is defined only against a candidate set; and two objects agreeing on every ranking can still differ as "worse" versus "forbidden"	**13.4%** of pairwise relations reverse when an irrelevant candidate is removed; the aggregate picks a forbidden response **26.1%** of the time
C4	*downstream behaviour* of a trained model	coherent, but requires Y, which is absent; and would measure the training stack as much as the standard	Y is not in the release and no arithmetic recovers it (§22)
C5	*mutual information* $I(N;G)$	counts dependence no decision uses: preserving wording while destroying polarity scores high	the rate–distortion repair works but its answer is fixed by the distortion choice, i.e. by C6's Q
C6	the *query-relative equivalence class* $[N]_Q$	—	the definition used here

Table 1: C1–C5 are not rivals to C6. Each is C6 at a particular Q: C1 at all text functions, C3 at ranking queries on the observed set, C5 at all measurable functions. Proposition 111 proves the choice of Q determines the answer, so the disagreement among C1–C5 is not empirical.

2.3 C6, stated

Definition 1 (Normative information)**.** Fix Q, the family of questions the downstream system must answer. Then $N \sim_Q N'$ iff $q(N,x) = q(N',x)$ for every $q \in Q$ and every x; the normative information in N is the class $[N]_Q$.

why C1–C5 each exercised this freedom silently. C6 makes it an input that must be published. That is the whole gain: not a better answer, an *answerable* question.

Theory claim. Declaring Q is not a preliminary to the measurement. **It is most of the result.** A framework reporting one percentage has chosen a Q and not disclosed it.

2.4 Where C6 lives formally, and what that buys

move	take Q to be decision problems rather than arbitrary queries — no loss of generality, since any query is posable as a decision
consequence	$N \sim_Q N'$ becomes Blackwell indistinguishability on that family
the quantity	*deficiency* $\delta_Q = \sup_{d \in Q} \left[R_{N'}(d) - R_N(d)\right]$, not a percentage
buys (i)	a **unit**: deficiency is in the units of the decision loss, so it is comparable to what the decision is worth
buys (ii)	a **worst case**: a supremum over the declared family, not an average over whichever queries were tried
buys (iii)	an **order**: Blackwell's relation is partial, so two standards can be incomparable, and saying so beats forcing a rank

2.5 Three source objects; merging them makes the question unanswerable

Z	the individual inputs $(N_1, \dots, N_n)$: who said what, who dissented, who vetoed	aggregation *necessarily* discards some of this (theorem 84)
G	$\mathsf{A}(Z)$, the collective norm under a rule A	loss here is a **social choice**, not an error — *if* A was declared
S	the authorised behaviour set after legitimate expert enrichment	loss here is either domain necessity or machine distortion

Proposition 2 (Without the layers, "defect" is undefined)**.** *A loss between Z and G is a defect only if unlicensed by* A. *If* A *is undeclared, no observed loss can be classified, and "the pipeline lost information" is indistinguishable from "the pipeline did what it said".*

Boundary. CoVal declares no A. At this site, authorised social choice and machine defect are **not separable**. Not a hard measurement problem — a missing line of documentation, and the cheapest thing a future release could fix.

2.6 The conservation law: four classes, not one number

why The deliverable of a theory of normative information is not a score. It is a statement of what may be compressed and what may not — the content is the *partition*, not the total.

class	distinctions	requirement on the pipeline
must be causally preserved	polarity; non-compensability; scope; exceptions	intervening must move behaviour, *in the predicted direction*
recoverable is enough	wording; non-load-bearing provenance; restatements	must be decodable; need not steer
licensed to drop	duplicates; entailed criteria; phrasing variance	only if the licence is declared *at the layer where it happens*
type errors when lost	veto → scalar; uncertainty → certainty; right → preference; expert constraint → public preference	a category change, not degradation; **no continuous metric detects it**

Table 2: Rows 2–3 are assertions. Row 1 is partly measured (§20). Row 4 is where theorem 90 sits.

2.7 Why row four is the deepest

the mechanism	a type change can leave the object arbitrarily close to the source in *every* continuous measure while answering a different question
the extreme case	a veto rendered as -10 and a strong dispreference rendered as -10 are the *same object*; there is nothing left to measure
where the difference lives	only in behaviour on candidate sets not yet seen — which is exactly theorem 90
consequence	similarity metrics are structurally unable to audit this class, and most alignment evaluation is similarity metrics

Theory claim. Normative information is not a quantity flowing down a pipe. It is a **partition of distinctions** into: must be causally preserved · must merely be recoverable · may be dropped under a declared licence · **cannot be represented at all in the chosen format**. The last class is invisible to every similarity measure, and it is where a scalar scale fails.

3 The object, in full

3.1 What was collected, field by field

field	index	content	measured property
prompt	p	a conversation	**968** joined; **18** of **986** rubric records never matched, so **1.8%** is invisible to every number here
responses	(p, r)	exactly four, letters A–D	fixes $k = 4$, hence the probe bound of two free parameters (theorem 74)
criterion text	(p, c)	"a good response should…"	**15,248** total, **4–39** per prompt, median **15**, median length **94** characters
weight	(p, c, a)	$-10..+10$	**102,147** ratings; exactly **one** is zero, so the scale is used strictly two-sided
ranking	(p, a)	`A>B>C=D`	three registers: `world` **100%**, `personal` **26.7%**, `unacceptable` **16.9%**
veto	(p, a)	set of "unacceptable"	indexed by *person*, not by criterion — the index mismatch of §13
`coval_core`	(p, k)	compiled standard, no weights	every item rewritten positive; disagrees with the full rubric on **29.4%** of prompts
satisfaction	(p, c, r)	**not released**	reconstructed with an LM judge; one-judge reliability **0.5497**, attenuation ceiling **0.741**
lineage	—	**absent**	compilation's three actions are confounded
A	—	**absent**	social choice and defect not separable
prompt id in rubrics	—	**absent**	the criterion→prompt index is reconstructed from message text

Table 3: The bottom block is what is *missing*, and it blocks more questions than the top block answers.

3.2 What it was for, and the three assumptions that carries

why Stating the design intent is not politeness. Each assumption below is a testable proposition, and testing the three of them *is* this paper.

	assumption	tested in	verdict
A1	what a person writes captures their values	§11, §19	**fails**: own criteria recover **39%** of own choice; the two routes disagree at **0.692** after full noise correction
A2	$-10..+10$ is an interval scale, addable across people	§11.2, §19	**unestablished and load-bearing**: worth **11.7%** of decisions; and cross-person addition requires an interpersonal stipulation the elicitation never made
A3	summing weighted criteria yields an executable standard	§13	**fails for one class**: no finite weight encodes a veto (theorem 91)

3.3 Why the design is nonetheless the right shape

why A catalogue of failures with no account of what the design got right is an attack, not an audit.

auditable by construction	criteria are text a human can read, so a disagreement can be *located*, unlike a preference-pair reward model
population-scale	the collective norm comes from a crowd rather than a handful of authors, which is the stated gap in constitutional approaches
two routes elicited	criteria *and* rankings from the same session — this redundancy is what made §19's central measurement possible at all
a veto field exists	most preference datasets have no way to say "forbidden"; this one does, which is why the type question is even askable here

4 The problems, catalogued

why

Each entry states what the problem is, what it was measured with, what it implies, and — where relevant — whether it is a property of this release or of the approach. The last column is what decides whether a fix is a bigger dataset or a different design.

	problem	measurement	implication	release or approach
P1	the two elicitations disagree beyond their own noise	disattenuated cosine **0.6924**	≈ **31%** of the write/choose gap is real, not measurement error	**approach**: any design eliciting both will face it
P2	the probe bound	rank 3, then S^2	**two** free parameters per person per prompt, whatever the criteria say	approach, unless k grows
P3	type collapse	**82.3%** of criteria carry no condition, exception, priority, hedge or prohibition	the questions "did compilation destroy the exception" are **empty** here	release: add fields
P4	the scale is used as interval, unvalidated	sign-only changes **11.7%** of decisions	every summed-criteria result inherits an unestablished assumption	release: validate or use paired comparison
P5	no declared aggregation rule	—	authorised social choice and machine defect **not separable**	release: one line of documentation
P6	no compilation lineage	29.4% disagreement, three actions	selection / rewriting / weight-dropping confounded	release: a receipt
P7	the instrument is a judge	reliability **0.5497**; **40%** of winners change with the reader; **8%** with page order	absolute claims are about the reader; comparative claims survive	approach: report both, always
P8	candidate-set dependence	**13.4%** IIA violations	measured "values" are partly a property of the four responses shown	approach: needs ≥ 2 candidate sets
P9	post-hoc contamination	judge scores criteria against the responses they were written about, **+0.0365**, $z = +46.2$	satisfaction and choice are two views of *one act*	release: response-blind elicitation
P10	manipulability	**3.9%** of annotator–prompt pairs, crudest misreport, a *lower bound*	the weights cannot be read as unconditionally sincere	approach: Gibbard–Satterthwaite guarantees it exists
P11	interpersonal comparability	—	every aggregate rests on a stipulation of equal stakes; the deficiency **0.1298** is a number about (data, scale)	approach: the oldest problem in social choice
P12	Y does not exist	—	any end-to-end number quoted to behaviour is fabricated	release, in principle

Table 4: Six are properties of this release and are fixable by collecting differently. Six are properties of the approach and are not.

5 What was attempted, and why each version failed

why

The failed constructions are load-bearing evidence, not autobiography: each one is a design somebody else will otherwise build, and the reason it fails is a property of the object rather than of the implementation.

	construction	what it was trying to capture	why it failed
V1	a single preservation percentage $N \to Y$	"how much survives" as one number	undefined without Q (proposition 111); and Y absent (P12)
V2	effective dimension of the induced norm	loss as *collapse of rank*	read 3.00 on every input *including its own null* — the probe bound (theorem 74) forced it. A check that cannot fail
V3	cosine between a person's rubric and their own ranking	alignment as preservation	the pipeline appears nowhere in it; and P9 makes the two sides views of one act
V4	a 19×23 operator $\times$ loss-shape detection matrix	which family sees which failure	**2** of the shapes are algebraically invariant to 1.8×10^{-14}; **5** of **8** operators were scalar multiples of one vector; the design's own rank was **2**
V5	eigenvalue rank of the operator set	operator distinctness as one integer	sits *inside* the span of three defensible nulls (12.1, 17.0, 18.0) with the observation at 13; the null *construction* moved it more than the data. Retired
V6	flip rate under perturbation	decision change as a probability	correct but a *coarsening*: discards $\mathbb{E}[\mathrm{Reg} \mid \mathrm{flip}]$, halving the measured separation between operators (lemma 115)
V7	normalised regret, swept target, cluster-bootstrapped, gauge-tested	the Blackwell deficiency on the declared decision family	**current**; exact on all eight known-truth quantities (§23)

Table 5: V2 and V5 are the instructive ones: both produced a confident number whose value was fixed by the construction rather than by the data.

5.1 The failure mode shared by V2 and V5, and by five withdrawn claims

shape	a verdict decided by a constant the author chose, rather than by a comparison against a range of defensible references
instances	one null instead of a null class · one reference instead of a reference class · a two-branch template for a three-outcome test · a 0.15 cutoff · a shape threshold in place of an outcome test · "rank 1" as a kill against a design of rank 2
why it is invisible	each constant is individually defensible; the failure is that it was *fixed before the outcome space was enumerated*
the repair, structural	every verdict is a printed comparison against the *range* of defensible references, never a boolean computed from a threshold

Boundary. This is reported because it is a measurable property of the methodology, not a confession. Five of six withdrawn claims share it; that is a higher concentration than any substantive cause, and it predicts where the next error will be.

6 The generalisation: what theorem 91 says beyond this dataset

why The theorem's premises never mention CoVal. Stating its actual scope is the difference between an audit of one dataset and a statement about a training stack.

premise (i)	the decision is arg max of a **scalar** score
premise (ii)	the **candidate set varies** between fitting and deployment
who satisfies both	every scalar reward model; every best-of-n sampler; every preference-trained ranker; every rubric summed to a number

Corollary 3 (Scalar reward cannot encode a constraint)**.** *Let a constraint be "never select from V". A scalar reward can express it only as a penalty M. By theorem 90 the penalty reproduces the constraint on a candidate set X iff $M > \Delta(X)$, and by theorem 91 $\sup_X \Delta(X) = \infty$. Hence no finite reward penalty enforces a constraint uniformly over candidate sets.*

what this is not	a claim that penalties never work. On any *fixed* candidate distribution a sufficient M exists and can be fitted
what this is	a claim that the sufficient M is a property of the **candidate distribution**, so it does not transfer — and the failure appears exactly where capability improves, because a better generator produces candidates that score higher on everything else
the measured instance	on **968** four-item sets the required penalty already reaches **2.65×** the entire elicitation scale
the architectural consequence	a constraint needs an **independent feasible set** — a classifier, a hard filter, a separate rejection head — not a term folded into the score
the falsifier	exhibit a bounded M reproducing constraint behaviour across candidate sets; theorem 91 forbids it for unbounded satisfaction, so a bounded-satisfaction variant with a stated bound would narrow the theorem rather than break it

Theory claim. The load-bearing result here is not about a dataset. **A scalar reward model cannot encode "never do X"; it can only encode "doing X costs a lot", and how much is a lot depends on what else was on the menu.** That is a type error, it is provable without any instrument, and it predicts that constraint violations concentrate exactly where the candidate distribution has shifted — which is where capability gains put it.

7 Every symbol, from arithmetic upward

This section defines everything the paper uses. It assumes arithmetic — adding, multiplying, and the idea of a number — and nothing else. A reader who is fluent may skip to §8; a reader who is not should be able to finish this section and then check every subsequent line.

7.1 Numbers, inequalities, intervals

$\mathbb{R}$ denotes the real numbers: everything on the number line, including fractions and irrationals like $\sqrt{2}$. $a < b$ means a lies to the left of b. $[a, b]$ is the set of all x with $a \le x \le b$ (endpoints included); (a, b) excludes the endpoints. $|x|$ is the *absolute value*: x if $x \ge 0$, and $-x$ otherwise. It is the distance from x to 0, so it is never negative.

One fact used repeatedly. Multiplying an inequality by a *positive* number preserves it; multiplying by a *negative* number **reverses** it. $2 < 3$ but $-2 > -3$. §8.11 depends on this at one specific step, and the step is marked there.

7.2 Functions

A *function* f from a set $\mathcal{A}$ to a set $\mathcal{B}$, written $f : \mathcal{A} \to \mathcal{B}$, assigns to each element of $\mathcal{A}$ exactly one element of $\mathcal{B}$. $\mathcal{A}$ is the *domain*, $\mathcal{B}$ the *codomain*. "Exactly one" is the whole content: a rule that sometimes gives two answers is not a function.

Definition 4 (Monotone)**.** f is *strictly increasing* if $x < y$ implies $f(x) < f(y)$. Such a function preserves order and nothing else: it can stretch and squeeze distances freely. §11.2 uses exactly this property to test whether the paper's central data are ordinal or not.

7.3 Sum and product notation

$$\sum_{i=1}^{n} x_i \;=\; x_1 + x_2 + \cdots + x_n, \qquad \prod_{i=1}^{n} x_i \;=\; x_1 \cdot x_2 \cdots x_n.$$

The letter i is a *dummy*: $\sum_{i=1}^{3} x_i$ and $\sum_{j=1}^{3} x_j$ are the same number. Three manipulation rules are used later and each is just ordinary arithmetic written compactly.

Rule 1 — pulling out a constant. $\sum_i (c\, x_i) = c \sum_i x_i$, because $cx_1 + cx_2 + \cdots = c(x_1 + x_2 + \cdots)$.

Rule 2 — splitting a sum. $\sum_i (x_i + y_i) = \sum_i x_i + \sum_i y_i$, because addition can be reordered.

Rule 3 — reordering, and why it is safe here. For a *finite* collection, adding the terms in any order gives the same total. Every sum in this paper is finite, so grouping terms differently — for instance grouping outcomes by the value a function takes on them, as in the proof of lemma 37 — never changes the answer. (For infinite sums this is false in general; the paper never needs one.)

Definition 5 (Factorial)**.** $n! := 1 \cdot 2 \cdot 3 \cdots n$, with $0! := 1$ by convention. It counts the number of ways to arrange n distinct objects in a row: n choices for the first, then $n - 1$, and so on. $3! = 6$, and the six orderings are listed explicitly in §16.

7.4 Powers, roots, logarithms

$x^2 := x \cdot x$; $\sqrt{x}$ is the non-negative number whose square is x; $x^{1/2} = \sqrt{x}$. The *logarithm* $\log x$ is the power to which the base must be raised to give x. Only two properties are used: $\log(ab) = \log a + \log b$ and log is strictly increasing. The second is why correlating log-transformed quantities in remark 73 is a statement about *proportional* rather than absolute co-movement.

7.5 Max, min, argmax, sup, inf

$\max_i x_i$ is the largest of the x_i; $\min_i x_i$ the smallest.

Definition 6 (argmax)**.** $\arg\max_i x_i$ is the *index* at which the maximum occurs, not the value. If $x = (4, 9, 2)$ then $\max = 9$ and $\arg\max = 2$. When two entries tie, the arg max is not unique; the paper's decision rules break ties by index order, and exact ties in continuous scores occur with probability zero in the data.

Definition 7 (Supremum)**.** sup is the least upper bound: the smallest number that is $\geq$ every element of a set. It coincides with max when the maximum is attained, and exists even when it is not. $\sup\{1 - \frac{1}{n} : n \geq 1\} = 1$ although 1 is never attained. theorem 91 states $\sup_X \Delta(X) = \infty$, meaning: for any bound you name, some candidate set exceeds it.

7.6 Sets, and the four operations used

$\mathcal{A} \cup \mathcal{B}$ (union) contains everything in either; $\mathcal{A} \cap \mathcal{B}$ (intersection) everything in both; $\mathcal{A} \subseteq \mathcal{B}$ (subset) means every element of $\mathcal{A}$ is in $\mathcal{B}$; $\varnothing$ is the empty set; $|\mathcal{A}|$ is the count of elements; $\mathcal{A} \setminus \mathcal{B}$ removes from $\mathcal{A}$ everything that is in $\mathcal{B}$.

Lemma 8 (Union bound, the pointwise version)**.** $\mathbf{1}\{x \in \bigcup_i \mathcal{A}_i\} \;\leq\; \sum_i \mathbf{1}\{x \in \mathcal{A}_i\}$.

Proof. If x is in none of the sets, both sides are 0. If x is in at least one, the left side is 1 and the right side is at least 1 because it counts the sets containing x. There is no third case. ◻

Theorem 59 is this lemma plus taking expectations, and nothing else.

7.7 Tuples and vectors

An *ordered tuple* $(x_1, \ldots, x_k)$ is a list in which position matters: $(1, 2) \neq (2, 1)$. The set of all such lists of k real numbers is written $\mathbb{R}^k$, and an element of it is a *vector*.

Throughout this paper $k = 4$ and the four positions are the four candidate responses A, B, C, D. So a vector such as $(5.1, 7.1, 5.7, 2.8)$ means: this norm scores response A at 5.1, B at 7.1, and so on.

Definition 9 (Addition and scaling)**.** $(x_1, \ldots, x_k) + (y_1, \ldots, y_k) := (x_1 + y_1, \ldots, x_k + y_k)$ and $c\,(x_1, \ldots, x_k) := (cx_1, \ldots, cx_k)$. Both act position by position.

Why this is the right operation here. A criterion with weight w and satisfaction profile $s = (s_A, s_B, s_C, s_D)$ contributes $w\,s$ to the score, and several criteria contribute their sum. That is exactly definition 71, and it is the only place vector arithmetic enters the construction.

7.8 The dot product, the norm, and the angle

Definition 10 (Dot product)**.** $x \cdot y := \sum_{i=1}^{k} x_i y_i$, a single number.

Definition 11 (Norm)**.** $\|x\| := \sqrt{x \cdot x} = \sqrt{\sum_i x_i^2}$.

Why the norm is the length. In two dimensions, $x = (x_1, x_2)$ is the corner of a right triangle with legs x_1 and x_2; Pythagoras gives hypotenuse $\sqrt{x_1^2 + x_2^2}$, which is $\|x\|$. In k dimensions the same formula is taken as the *definition* of length, and it behaves as length should: $\|cx\| = |c|\,\|x\|$, and $\|x\| = 0$ only when every coordinate is 0.

Theorem 12 (The dot product measures angle)**.** *For non-zero x, y,*

$$\cos\theta \;=\; \frac{x \cdot y}{\|x\|\,\|y\|},$$

where θ is the angle between x and y.

Proof. S1 The law of cosines for the triangle with sides $\|x\|$, $\|y\|$ and $\|x - y\|$ states $\|x - y\|^2 = \|x\|^2 + \|y\|^2 - 2\|x\|\|y\|\cos\theta$.

S2 Expand the left side using the definition:

$$\|x-y\|^2 = \sum_i (x_i - y_i)^2 = \sum_i x_i^2 - 2\sum_i x_i y_i + \sum_i y_i^2 = \|x\|^2 - 2(x\cdot y) + \|y\|^2.$$

S3 Set the two expressions equal. The terms $\|x\|^2$ and $\|y\|^2$ cancel from both sides, leaving $-2(x\cdot y) = -2\|x\|\|y\|\cos\theta$.

S4 Divide by $-2\|x\|\|y\|$.

□

Definition 13 (Cosine similarity)**.** $\cos(x, y) := \dfrac{x \cdot y}{\|x\|\|y\|}$, a number in $[-1, 1]$. $+1$ means same direction, 0 perpendicular, -1 opposite.

Remark 14 (The single most important consequence for this paper). Cosine similarity is *blind to length*: $\cos(x, y) = \cos(2x, y)$ for any scaling. That is precisely why it is the right tool once remark 73 establishes that the length of N_i measures how many criteria a person happened to rate rather than anything normative.

A worked cosine. $x = (1, 0)$, $y = (1, 1)$. Then $x \cdot y = 1$, $\|x\| = 1$, $\|y\| = \sqrt{2} = 1.4142$, so $\cos = 1/1.4142 = 0.7071$, which is $\cos 45°$. Correct: $(1, 1)$ points at 45 degrees to $(1, 0)$.

7.9 Centring, and what it removes

Definition 15 (Centred vector)**.** $\tilde{x} := x - \bar{x}\,\mathbf{1}$ where $\bar{x} = \frac{1}{k}\sum_i x_i$ and $\mathbf{1} = (1, 1, \ldots, 1)$. Equivalently, subtract the average from every entry.

Two facts, both immediate. $\sum_i \tilde{x}_i = \sum_i x_i - k\bar{x} = 0$; and if $y = x + c\mathbf{1}$ then $\tilde{y} = \tilde{x}$, because the shift moves the average by exactly c as well. *Centring erases the common level and keeps only the contrasts*, which is exactly what the decision uses.

7.10 Matrices, rank, and the one matrix this paper needs

A *matrix* is a rectangular array of numbers; M_{ij} is the entry in row i, column j. A matrix acts on a vector by $(Mx)_i := \sum_j M_{ij}x_j$: each output coordinate is a weighted sum of the input coordinates.

Definition 16 (Linear independence, span, rank)**.** Vectors $v_1, \ldots, v_m$ are *linearly independent* if the only way to make $\sum_j c_j v_j = 0$ is to take every $c_j = 0$ — none of them is redundant. Their *span* is the set of all $\sum_j c_j v_j$. The *rank* of a matrix is the number of linearly independent vectors among its outputs: the dimension of the space it can reach.

Lemma 17 (The centring matrix and its rank)**.** *Let $C := I - \frac{1}{k}\mathbf{1}\mathbf{1}^\top$, so that $Cx = \tilde{x}$. Then C is a projection ($C^2 = C$) and* $\operatorname{rank}(C) = k - 1$.

Proof. S1 $(\mathbf{1}\mathbf{1}^\top x)_i = \sum_j x_j$ for every i, so $\frac{1}{k}\mathbf{1}\mathbf{1}^\top x = \bar{x}\mathbf{1}$, and $Cx = x - \bar{x}\mathbf{1}$.

S2 Centring an already-centred vector changes nothing, since its average is 0. So $C(Cx) = Cx$, i.e. $C^2 = C$.

S3 $C\mathbf{1} = \mathbf{1} - \mathbf{1} = 0$, so the one-dimensional space of constant vectors is sent to zero.

S4 If $Cx = 0$ then $x = \bar{x}\mathbf{1}$, i.e. x is constant. So the set sent to zero is *exactly* the constants, of dimension 1.

S5 The input space has dimension k; a one-dimensional subspace is destroyed; the image therefore has dimension $k - 1$. For $k = 4$: rank 3.

□

Lemma 17 is the entire content of the first half of theorem 74.

7.11 Equivalence, quotients, and the sphere

Definition 18 (Equivalence relation)**.** A relation $\sim$ that is reflexive ($x \sim x$), symmetric ($x \sim y \Rightarrow y \sim x$) and transitive ($x \sim y, y \sim z \Rightarrow x \sim z$). It partitions a set into disjoint *equivalence classes*: $[x] := \{y : y \sim x\}$.

The two used here. (i) $u \sim u + c\mathbf{1}$: utilities differing by a constant, which decide identically. (ii) $u \sim \lambda u$ for $\lambda > 0$: utilities differing by positive scale, which order identically.

Definition 19 (Quotient)**.** The set of equivalence classes, written $\mathcal{A}/\!\sim$. Its dimension is the dimension of $\mathcal{A}$ minus the dimension of what was collapsed.

Definition 20 (Sphere)**.** $S^m := \{x \in \mathbb{R}^{m+1} : \|x\| = 1\}$, the set of unit-length vectors in $\mathbb{R}^{m+1}$. It has dimension m: one constraint removes one degree of freedom. S^1 is a circle in the plane; S^2 is the surface of a ball in three dimensions.

Putting the two together, which is theorem 74. Starting from $\mathbb{R}^4$: quotient by constants leaves dimension 3; the unit-length representatives of that 3-dimensional space form S^2, of dimension 2. Hence "two free parameters".

7.12 Order statistics: median, quantile, percentile

Sort the values as $x_{(1)} \le x_{(2)} \le \cdots \le x_{(n)}$.

Definition 21 (Median)**.** The middle value: $x_{((n+1)/2)}$ if n is odd, and the average of $x_{(n/2)}$ and $x_{(n/2+1)}$ if n is even. For $(2, 4, 4, 4, 6)$ it is 4; for $(2, 4, 4, 6)$ it is $(4 + 4)/2 = 4$.

Definition 22 (Quantile)**.** The q-quantile is the value below which a fraction q of the data lies. The median is the 0.5-quantile; p_{25} and p_{75} are the 0.25- and 0.75-quantiles (*quartiles*). "$p_{95} = 0.800$" in §20 means 95% of prompts have a relative margin below 0.800.

Remark 23 (Why the paper reports quantiles beside means). A mean can be moved arbitrarily by one extreme value; a median cannot. Reporting both, as section 13.2 does with median 3.64 and maximum 26.54, exposes a skew that a mean alone would hide — and in that table the *maximum* is the load-bearing number, because theorem 91 is about the worst case.

7.13 Rank correlation (Spearman)

Definition 24 (Spearman ρ)**.** Replace each value by its rank (smallest = 1, next = 2, ...) in its own list, then compute the ordinary correlation of the ranks.

Why the paper uses it for the gauge test. It responds only to *order*, so it answers "do the two instruments rank the operators the same way" without being distracted by their different scales — which is the exact question §15.3 asks. A Spearman of +1.000 means identical ordering; 0 means unrelated ordering.

Worked. Instrument 1 gives $(0.116, 0.051, 0.042, 0.023)$, ranks $(4, 3, 2, 1)$. Instrument 2 gives $(0.139, 0.061, 0.035, 0.025)$, ranks $(4, 3, 2, 1)$. Identical ranks, so $\rho = +1.000$, even though no value matches.

7.14 Bits, and what quantisation means

Definition 25 (Bit)**.** One *bit* distinguishes two possibilities. b bits distinguish 2^b: one bit gives 2 levels, two bits 4, three bits 8, eight bits 256.

Definition 26 (Quantisation to b bits)**.** Given values in $[\ell, h]$, replace each x by the nearest of the 2^b evenly spaced levels $\ell + \frac{j}{2^b-1}(h - \ell)$, $j = 0, \ldots, 2^b - 1$.

Worked, at one bit. With weights ranging over $[-2, 8]$, one bit leaves the two levels $\{-2, 8\}$: every negative weight becomes -2 and every positive one 8. Only the *sign* survives. §14 reports that this reproduces 89.8% of the shipped decisions.

7.15 Min–max normalisation

Definition 27. $\hat{x}_i := \dfrac{x_i - \min_j x_j}{\max_j x_j - \min_j x_j} \in [0, 1]$, with $\hat{x} = 0$ at the worst entry and 1 at the best.

Why §12 uses it, and what it assumes. Comparing utility across people requires putting them on a common scale, and nothing in the elicitation supplies one. Min–max makes every person's best option worth 1 and worst worth 0 *by construction*, which is a stipulation, not a measurement. It is stated as such: the Blackwell deficiency of 0.1298 is a number relative to that normalisation, and a different interpersonal scale would give a different figure.

7.16 Conditional probability and conditional expectation

Definition 28 (Conditional probability)**.** $\mathbb{P}(E \mid F) := \dfrac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)}$ when $\mathbb{P}(F) > 0$: restrict attention to the outcomes in F and renormalise so they sum to 1.

Definition 29 (Conditional expectation)**.** $\mathbb{E}[X \mid F] := \dfrac{\sum_{\omega \in F} X(\omega)\mathbb{P}(\omega)}{\mathbb{P}(F)}$, the average of X over F alone.

Lemma 30 (Law of total expectation, the two-event case)**.** *For an event F with complement F^c,*

$$\mathbb{E}[X] \;=\; \mathbb{E}[X \mid F]\,\mathbb{P}(F) \;+\; \mathbb{E}[X \mid F^c]\,\mathbb{P}(F^c).$$

Proof. Split the defining sum by whether the outcome lies in F:

$$\mathbb{E}[X] = \sum_{\omega \in F} X(\omega)\mathbb{P}(\omega) + \sum_{\omega \in F^c} X(\omega)\mathbb{P}(\omega).$$

Multiply and divide the first sum by $\mathbb{P}(F)$ and the second by $\mathbb{P}(F^c)$, then apply the definition above to each. (If $\mathbb{P}(F) = 0$ the term is absent and the identity is trivial.) □

Lemma 30 with $F = \{\text{the decision flipped}\}$ and $X = \mathrm{Reg}$ is exactly lemma 115, since $\mathrm{Reg} = 0$ on F^c so the second term vanishes.

7.17 Independent and identically distributed

Identically distributed means every observation is drawn from the same $\mathbb{P}$. *Independent* means no observation carries information about another. Written *iid*. §8.14 exists because in this data the observations are identically distributed but **not** independent, and only the second half fails.

7.18 Markov kernels

Definition 31 (Markov kernel)**.** A rule K that, given an input z, produces a *probability distribution* over outputs, written $K(\cdot \mid z)$. A deterministic function is the special case that puts all the probability on one output, written $\delta_{f(z)}$.

Why the paper needs it. Theorem 99 says a garbling cannot help. Aggregation is deterministic, hence a garbling with the degenerate kernel $\delta_{\mathsf{A}(z)}$; the theorem then applies verbatim without any probabilistic aggregation being involved.

7.19 Integrals, only as much as Φ requires

The paper uses the normal distribution function $\Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-t^2/2}\,dt$. The integral sign means *area under the curve to the left of* z. The curve $\frac{1}{\sqrt{2\pi}} e^{-t^2/2}$ is the familiar bell; its total area is 1, so $\Phi(z)$ is the fraction of the bell lying left of z.

Definition 32 (Quantile of the normal)**.** z_q is the point with $\Phi(z_q) = q$. The three used are $z_{0.975} = 1.9600$, $z_{0.80} = 0.8416$, $z_{0.99} = 2.3263$. By the symmetry of the bell about 0, $z_{1-q} = -z_q$; this symmetry is the step used in theorem 54.

Remark 33 (No calculus is needed beyond this). The paper never differentiates or integrates anything. Φ enters only as a table of three numbers, and every use of it is an instance of "how far out in the bell is this".

7.20 Sigmoid and logit, for the instrument

The judge produces a number λ (a difference of log-probabilities) and it is mapped to $(0, 1)$ by the *sigmoid* $\sigma(\lambda) = 1/(1 + e^{-\lambda})$, which is strictly increasing with $\sigma(0) = 0.5$.

Boundary. $\sigma(\lambda)$ lies in $(0, 1)$ and is therefore easily mistaken for a probability. It is not one. It is a monotone re-expression of a model's internal score, calibrated against nothing. Every use of S in this paper carries that caveat, and §15 measures its consequence rather than assuming it away.

7.21 A table of every symbol

symbol	meaning	first defined
$\mathbf{1}\{P\}$	indicator, 1 if P true	§8
$\sum$, $\prod$, $n!$	sum, product, factorial	§7
arg max	the index of the maximum	§7
sup, inf	least upper / greatest lower bound	§7
$x \cdot y$, $\|x\|$	dot product, length	§7
$\cos(x, y)$	cosine similarity	§7
$\tilde{x}$	centred vector	§7
C	centring matrix, rank $k - 1$	lemma 17
S^m	unit sphere, dimension m	§7
$\mathbb{P}$, $\mathbb{E}$, Var, Cov, Corr	probability, mean, variance, covariance, correlation	§8
se	standard error = $\mathrm{sd}/\sqrt{n}$	§8.8
s^2	sample variance, divisor $n - 1$	lemma 50
Φ, z_q	normal cdf and its quantiles	§7
ρ	intraclass correlation *or* Spearman	§8.14, §7
DEFF	design effect $1 + (R - 1)\rho$	theorem 57
MDE	minimum detectable effect	theorem 54
rel	reliability	§8.18
$\mathcal{X}$, X, k	response space, probe set, its size (= 4)	§10
S, W, V, Π	satisfaction, weight, veto set, ranking	definition 69
Y_p, N_i	collective and individual induced scores	definition 71, definition 72
A, G	aggregation rule and its output	equation (4)
Reg	normalised regret	definition 113
π	flip rate	§20
m	decision margin	proposition 118
Δ	the penalty a veto requires	theorem 90
ϕ_i	Shapley influence	§21
$v(S)$	characteristic function of the coalition game	§21
$K(\cdot \mid z)$	Markov kernel	§7
$\sigma(\lambda)$	sigmoid	§7

Table 6: Every symbol used in the paper. ρ is overloaded by convention; the two uses never appear in the same expression and are disambiguated at each site.

8 The statistics, and what each concept actually is

why A definition is a rule for writing a symbol. It does not say why the object exists, what it feels like, or what breaks without it. Each concept below is therefore given the problem it was invented to solve, a picture, the definition, one worked number, and the misreading that is standard in this literature.

8.1 Probability

problem We need to talk about "how often" before we have seen everything. Without a rule, "often" is a word; with one, it is arithmetic that composes.

picture A finite bag of outcomes with weights that sum to 1. An event is a handful of outcomes; its probability is the weight of the handful. Nothing is mysterious or infinite — every probability in this paper is a fraction of a finite set.

Definition 34 (Probability space)**.** A finite set Ω of outcomes with $\mathbb{P}(\omega) \geq 0$ summing to 1; $\mathbb{P}(E) := \sum_{\omega \in E} \mathbb{P}(\omega)$ for $E \subseteq \Omega$.

one number 968 prompts, 42 of which have a binding veto: the probability that a randomly drawn prompt has one is $42/968 = 0.0434$.

misread as "Probability" as a property of a single case. It is a property of a *procedure for drawing*. "This prompt has probability 0.043" is not a sentence; "a prompt drawn uniformly from these 968" is.

8.2 The indicator, which turns yes/no into arithmetic

problem Most questions here are yes/no — did the decision change, is the veto binding. Yes/no cannot be averaged, added, or given an interval.

picture Write 1 for yes and 0 for no. Now the *average* of the answers is exactly the *fraction* of yeses, so every tool built for averages applies unchanged to proportions. This one move is why the whole paper needs only one statistical toolkit.

$$\frac{1}{n}\sum_{i=1}^{n}\mathbf{1}\{P_i\} = \frac{\#\{i : P_i \text{ true}\}}{n}. \tag{1}$$

one number Flip / no-flip over 968 prompts, coded 1/0: the average is 0.051, and that number is simultaneously "the mean of the indicator" and "5.1% of prompts flipped". Same object.

8.3 Random variable

problem We want to attach numbers to outcomes and then reason about the numbers.

picture A rule that reads a number off whatever happened. It is neither random nor a variable — the randomness is in which outcome occurs, and the rule is fixed. "X is random" is shorthand for "the input to X is drawn".

Definition 35. $X : \Omega \to \mathbb{R}$, and $\mathbb{P}(X = x) := \mathbb{P}(\{\omega : X(\omega) = x\})$.

8.4 Expectation

problem One number summarising where a quantity sits, that does not depend on how many times you looked.

picture Take the long-run average. If value x shows up a fraction $\mathbb{P}(X = x)$ of the time, then in n draws it contributes $x \cdot n\mathbb{P}(X = x)$; divide by n and the n cancels. What remains does not depend on n, so it is a property of the distribution rather than of the experiment. *That* is why it is worth a name.

Definition 36. $\mathbb{E}[X] := \sum_x x\,\mathbb{P}(X = x)$.

Lemma 37 (Linearity)**.** $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$, *with* ***no independence assumption****.*

Proof. Sum over outcomes rather than values — legitimate because grouping a finite sum differently does not change it. Then $\sum_\omega (aX(\omega) + bY(\omega))\mathbb{P}(\omega) = a\sum_\omega X(\omega)\mathbb{P}(\omega) + b\sum_\omega Y(\omega)\mathbb{P}(\omega)$. □

picture Linearity is the only tool in the paper that never needs a condition. Every later step where an assumption would be contentious is routed through it deliberately.

Corollary 38 (The bridge)**.** $\mathbb{E}[\mathbf{1}\{P\}] = \mathbb{P}(P)$*: a measured proportion estimates a probability.*

misread as "Expectation" as what you expect to see. A fair die has expectation 3.5 and never shows it. It is a centre of mass, not a prediction.

8.5 Variance and standard deviation

problem Two datasets can have the same average and be completely different. We need a number for *spread*, and the obvious candidate fails.

picture Try averaging the deviations from the mean: it is *exactly zero* for every dataset, because positives cancel negatives. So the sign must be removed. Squaring does it, and squaring has one property absolute value lacks: squared spreads *add* when quantities are independent, which is what makes the spread of an average computable at all (equation (2)).

Definition 39. $\mathrm{Var}(X) := \mathbb{E}[(X - \mu)^2]$; $\mathrm{sd}(X) := \sqrt{\mathrm{Var}(X)}$.

picture The standard deviation is in the same units as the data — it is a *typical distance from the middle*. The variance is in squared units and is an intermediate quantity, which is why every interval in this paper is written with the standard deviation and never the variance.

Lemma 40 (Computational form)**.** $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

Proof. $\mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] = \mathbb{E}[X^2] - 2\mu^2 + \mu^2$ by lemma 37. □

Corollary 41 (Variance of a yes/no quantity)**.** *For* $X = \mathbf{1}\{P\}$ *with* $\mathbb{P}(P) = \pi$*: since* $0^2 = 0$ *and* $1^2 = 1$ *we have* $X^2 = X$*, so* $\mathrm{Var}(X) = \pi - \pi^2 = \pi(1-\pi)$.

one number $\pi = 0.5 \Rightarrow \mathrm{Var} = 0.25$; $\pi = 0.05 \Rightarrow \mathrm{Var} = 0.0475$. A rare event is intrinsically *easier* to pin down than a coin flip, which is why the 5% flip rates in table 16 have narrow intervals at modest n.

Lemma 42 (Scaling)**.** $\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$*: shifting everything does not change spread; stretching by* a *stretches spread by* $|a|$.

Proof. The deviation of $aX+b$ from its mean $a\mu+b$ is $a(X-\mu)$; square and take expectations. □

grounded Two prompts, both with mean satisfaction 0.5 across the four responses. Prompt A: $(0.50, 0.50, 0.50, 0.50)$ — variance 0, and *no rubric can decide anything here*, because every response is equally satisfying. Prompt B: $(0.95, 0.55, 0.45, 0.05)$ — variance 0.101, and the decision is easy. Same mean, opposite decidability. **The variance of the satisfaction vector is what makes a criterion useful at all**, which is why a criterion that is constant across responses contributes exactly nothing (measured at 0.0% in table 16).

grounded Second use, at a different level. Across 968 prompts the flip indicator has $\pi = 0.051$, so by corollary 41 its variance is 0.0484 — and that single number is what the entire interval [4.4%, 5.8%] is computed from. Nothing else about the data enters.

misread as A large variance as "bad data". It is a fact about the quantity, not a defect. What is bad is a large variance *of the estimate*, which is a different object (§8.8).

8.6 Independence, covariance, correlation

problem We want to say "these two move together" without saying how strongly in arbitrary units.

Definition 43 (Independence)**.** $\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y)$ for all x, y: knowing one tells you nothing about the other.

picture **Covariance:** multiply the two deviations and average. When both are above their means the product is positive; when they move oppositely it is negative; the average is therefore positive for co-movement. Its size is in the product of the two units — "weight-points times satisfaction" — which is uninterpretable, and that is the whole reason correlation exists.

Definition 44. $\mathrm{Cov}(X,Y) := \mathbb{E}[(X-\mu_X)(Y-\mu_Y)]$; note $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$.

Lemma 45 (Variance of a sum)**.** $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y)$*; under independence the cross term is* 0.

Proof. $(D_X + D_Y)^2 = D_X^2 + 2D_XD_Y + D_Y^2$ with $D_X := X - \mu_X$; take expectations. Under independence $\mathbb{E}[D_XD_Y] = \mathbb{E}[D_X]\mathbb{E}[D_Y] = 0$. □

Theorem 46 (Cauchy–Schwarz)**.** $|\mathrm{Cov}(X,Y)| \le \mathrm{sd}(X)\,\mathrm{sd}(Y)$.

Proof. For any t, $0 \le \mathbb{E}[(D_X + tD_Y)^2] = \mathrm{Var}(X) + 2t\,\mathrm{Cov}(X,Y) + t^2\,\mathrm{Var}(Y)$. A quadratic that is never negative has discriminant ≤ 0: $4\,\mathrm{Cov}^2 - 4\,\mathrm{Var}(X)\,\mathrm{Var}(Y) \le 0$. □

Definition 47 (Correlation)**.** $r := \mathrm{Cov}(X,Y)/(\mathrm{sd}(X)\,\mathrm{sd}(Y)) \in [-1, 1]$ by theorem 46.

picture Correlation is covariance with the units divided out. r answers: if one quantity is one standard deviation above its mean, how many standard deviations above is the other, on average? $r = 0.3$ means "about three tenths of one".

one number $r = +0.304$ between Shapley influence and alignment (§21). Read: a person one standard deviation more aligned than average is, on average, 0.30 standard deviations more influential. That is a real relationship and a weak one, which is exactly the finding.

grounded Concretely, in the three-person example of §16: annotator 2 has alignment 0.9313 — nearly perfect agreement with the collective outcome — and Shapley influence 0.3333, less than half of annotator 1's 0.8333 at a *lower* alignment of 0.7858. Two people, and the two measures already rank them differently. Across 1,865 real pairs that shows up as $r = +0.304$: the two quantities move together, weakly, and one cannot be substituted for the other.

grounded What a *high* correlation would have meant. Had r come out ≈ 1.0, the pre-registered conclusion was that influence and alignment are the same thing and sixty rounds of correlational measurement had been fine. The number decided between two research programmes, which is why it was registered before running.

misread as Three, all common. (i) r as a percentage — it is not; only r^2 has a share reading, and only under a linear model. (ii) $r = 0$ as "unrelated" — covariance sees only *linear* co-movement; a quantity can be perfectly determined by another with $r = 0$. (iii) r as comparable across studies — it is scaled by the spreads present in *that* sample, so a restricted range shrinks it mechanically.

8.7 Estimator, estimand, bias

problem We never see $\mathbb{P}$. We see n numbers and must guess.

picture The *estimand* is the fixed unknown we want. The *estimator* is a recipe applied to data. The estimator wobbles from study to study; the estimand does not move at all. Almost every misreading in applied statistics is a confusion about which of the two is doing the moving.

Definition 48. $\mathrm{Bias}(\hat{\theta}) := \mathbb{E}[\hat{\theta}] - \theta$.

picture Unbiased means: right *on average across repetitions of the whole study*. It says nothing about being close in any one study. A recipe can be unbiased and wildly variable, or biased and reliably close.

misread as "Unbiased" as "accurate". They are different axes: bias is where the centre sits, variance is how far it scatters.

8.8 The standard error

problem Our estimate came from one sample. How far might it be from the truth, purely because a different sample would have given a different number?

picture Averaging cancels noise. If each observation is off by a typical amount σ, and the errors are independent, they partly cancel — but only partly, and the amount of cancellation is $\sqrt{n}$, not n, because errors add in *squares* (lemma 45) and lengths are square roots of squares. The standard error is the typical distance between your estimate and the truth.

$$\begin{aligned} \mathbb{E}[\bar{X}] &= \mu && \text{by lemma 37} \\ \mathrm{Var}(\bar{X}) &= \frac{1}{n^2}\sum_i \mathrm{Var}(X_i) = \frac{\sigma^2}{n} && \text{by lemma 42, lemma 45} \\ \mathrm{se}(\bar{X}) &= \sigma/\sqrt{n} \end{aligned} \tag{2}$$

one number $\sigma = 0.124$, $n = 968$: se $= 0.124/31.1 = 0.0040$. So a flip rate measured at 5.1% would typically land within about 0.4 points of the truth — and to halve that to 0.2 points you need *four times* the prompts, not twice.

picture The $\sqrt{n}$ is the entire economics of measurement. It is why §20 can say "1,213 prompts for 1pp resolution" and mean it as a budget, not a wish.

grounded The same measurement at three sample sizes, holding $\sigma = 0.124$ fixed. $n = 10$: se $= 0.039$, interval ± 7.7 points — a 5.1% rate is indistinguishable from 0% or from 12%, so the study says nothing. $n = 100$: se $= 0.0124$, ± 2.4 points — now 5% is distinguishable from 0 but not from 8%. $n = 968$: se $= 0.0040$, ± 0.8 points — the actual study. **Same phenomenon, same estimate, three different claims**, and only the third supports the sentence "deleting flips more often than doubling" (a 0.93 point gap).

grounded And the reason $\sqrt{n}$ hurts. To resolve 0.5 points instead of 1 point, corollary 55 demands $n = (2.8016 \times 0.124/0.005)^2 = 4{,}827$ prompts — five times the release, for twice the resolution. This is why §28 states a sample-size requirement rather than a wish.

misread as The standard error as the spread of the *data*. It is the spread of the *estimate*, which is $\sqrt{n}$ times smaller. Data spread does not shrink with more data; estimate spread does.

Remark 49 (Independence is doing real work here). If observations are positively correlated, lemma 45 adds covariance terms and the true spread is *larger* than σ^2/n. Using equation (2) anyway understates uncertainty — §8.14 measures exactly how much, because this error occurred in this work.

8.9 Why $n-1$

problem To use $\mathrm{se} = \sigma/\sqrt{n}$ we need σ, which we also do not know. Estimating it from the same data introduces a specific, correctable error.

picture Deviations are measured from $\bar{X}$, which was computed *from the same points* and therefore sits closer to them than the true μ does. So the squared deviations come out systematically too small. Dividing by $n-1$ instead of n inflates them by exactly the right amount.

Lemma 50. $\mathbb{E}\left[\sum_i (X_i - \bar{X})^2\right] = (n-1)\sigma^2$.

Proof. $\sum_i (X_i - \bar{X})^2 = \sum_i (X_i - \mu)^2 - n(\bar{X} - \mu)^2$ after expanding around μ and using $\sum_i (X_i - \mu) = n(\bar{X} - \mu)$. Take expectations: the first term is $n\sigma^2$, the second is $n \cdot \sigma^2/n = \sigma^2$ by equation (2). □

picture "One degree of freedom was spent locating the centre." With $n = 2$ points, the mean sits exactly between them and the deviations carry only *one* piece of independent information about spread, not two — hence $n - 1 = 1$.

8.10 The central limit theorem

problem To turn se into an interval we need to know the *shape* of the estimate's wobble, and the data here are $0/1$ — about as far from a bell as data gets.

picture Averaging manufactures a bell. Whatever the individual observations look like, their average is approximately normally distributed once n is moderate, because an average is a sum of many small independent contributions and sums of many small independent things pile up in the middle. The data do not become normal; the *average* does.

Theorem 51 (Stated, not proved)**.** *For iid X_i with mean μ and finite variance, $(\bar{X}-\mu)/(\sigma/\sqrt{n})$ approaches the standard normal.*

licenses	normal-based intervals for *averages*, including rates built from $0/1$ indicators
does not say	that the data are normal — they are not, and it does not matter
fails when	observations are dependent (§8.14); n small and the distribution skewed; or the quantity is not an average, e.g. a correlation — which is why §8.15 resamples instead

Definition 52 (The three numbers used)**.** z_q is the point with a fraction q of the bell to its left: $z_{0.975} = 1.9600$, $z_{0.80} = 0.8416$, $z_{0.99} = 2.3263$. The bell is symmetric about 0, so $z_{1-q} = -z_q$.

8.11 Confidence intervals

problem A single number with no indication of its precision is unusable: 5.1% from ten prompts and 5.1% from a thousand are different claims.

picture Build a net around the estimate wide enough that, if you repeated the whole study forever, 95% of the nets would catch the fixed truth. **The net moves; the truth does not.** Everything confusing about intervals comes from imagining it the other way round.

- S1 From theorem 51, with probability 0.95: $-1.96 \le (\bar{X} - \mu)/\mathrm{se} \le 1.96$.
- S2 Multiply through by $\mathrm{se} > 0$; inequalities are preserved because the multiplier is positive: $-1.96\ \mathrm{se} \le \bar{X} - \mu \le 1.96\ \mathrm{se}$.
- S3 Subtract $\bar{X}$, then multiply by -1 — which **reverses** both inequalities: $\bar{X} - 1.96\ \mathrm{se} \le \mu \le \bar{X} + 1.96\ \mathrm{se}$.

one number 5.1% with se = 0.37pp gives [4.4%, 5.8%]. Read: a study like this one, repeated, would produce intervals catching the truth 19 times in 20.

grounded Two studies reporting the identical estimate 5.1%. Study 1: $n = 25$, interval [0.0%, 10.9%] — compatible with no effect at all. Study 2: $n = 968$, interval [4.4%, 5.8%] — incompatible with anything below 4 points. **The point estimate carries none of this.** Quoting 5.1% without the interval hides the entire difference between a result and a rumour.

misread as Three, in increasing order of consequence. (i) "There is a 95% chance μ is in this interval" — μ is a constant; it is in or it is not. (ii) "95% of future observations fall here" — that is a prediction interval and is far wider. (iii) "The interval excludes zero, so the effect is large" — exclusion is about *precision*, not size. A trivial effect measured well excludes zero; §8.13 separates the two questions.

8.12 Tests, z, and p

problem We want a rule for "this is more than nothing" that cannot be tuned after seeing the answer.

picture Assume the boring world (the effect is zero). Ask how surprising the data would be *in that world*. If very surprising, the boring world is a poor description. Note what this does *not* do: it never evaluates the interesting world, and it never assigns a probability to either.

Definition 53. $z := \hat{\theta}/\mathrm{se}$, the estimate measured in units of its own uncertainty; $p := \mathbb{P}(|Z| \geq |z| \mid H_0)$.

one number $z = +3.3$ for the paired delete-versus-double difference. Read: the observed gap is 3.3 of its own standard errors away from zero.

misread as Three, all false. (i) p is not $\mathbb{P}(H_0 \mid \text{data})$ — it is conditioned the other way. (ii) p is not the probability the result is a fluke. (iii) $1 - p$ is not the probability the effect is real. A p-value is a statement about *data under an assumption*, never about the assumption given data. This paper mostly reports z and intervals, because a p-value collapses size and precision into one number and they must stay separate.

8.13 Power, and the minimum detectable effect

problem A null result is uninterpretable on its own. "We found nothing" and "we could not have found anything" produce the same output.

picture Two bells: one centred at zero (the boring world), one centred at the true effect. The test rejects when the estimate lands past a threshold in the first bell's tail. *Power* is how much of the *second* bell sits past that threshold. If the two bells overlap heavily — a small effect, or a large se — most of the second bell is on the wrong side and the study will miss a real effect most of the time. The MDE is the separation at which 80% of the second bell finally clears.

Theorem 54. $\mathrm{MDE} = (z_{1-\alpha/2} + z_{1-\beta})\ \mathrm{se}$.

Proof. Reject when $\hat{\theta} \geq z_{1-\alpha/2}\,\mathrm{se}$. If the truth is δ, then $(\hat{\theta}-\delta)/\mathrm{se}$ is standard normal, so rejection requires $(\hat{\theta}-\delta)/\mathrm{se} \geq z_{1-\alpha/2} - \delta/\mathrm{se}$. That has probability $1 - \beta$ exactly when the right side equals $-z_{1-\beta}$, by symmetry of the bell. Solve for δ. □

one number se = 0.37pp gives MDE = 2.8016×0.37 = 1.05pp. Read: this design would reliably notice a true flip rate difference of about one point and would usually miss half a point.

Corollary 55 (Sample size)**.** *$n = (2.8016\,\sigma/\delta)^2$ to detect δ.*

grounded A real case from this programme, and it explains a retraction. A round claimed "the compiled standard respects vetoes better than a human peer", with a measured gap of 1.72 points; the claim was later withdrawn after it failed in 7 of 12 held-out halves, and nobody asked why. Computing the MDE from the round's own published interval: se = 0.0069, so MDE = 2.8016×0.0069 = 1.93 points. **The design's minimum detectable effect was larger than the effect it was measuring.** The full-sample interval barely excluding zero was luck; halving the data removed the luck. The held-out sweep was rediscovering a power problem twelve times, and two lines of arithmetic — available from numbers already written down — would have said so at the start.

grounded The same computation used forwards. The paired delete-versus-double difference is +0.93 points with MDE = 0.79. Effect exceeds MDE, but only by a factor of 1.2: the finding is real and its margin is thin, and that is stated in table 16 rather than buried.

misread as A null as evidence of absence. If MDE exceeds the effect the null was meant to rule out, the null is *silence*. Every null in §23 carries its MDE for this reason.

8.14 Clustering, ICC, and the design effect

problem We have 19,360 rows but only 968 prompts, each perturbed 20 times. Treating the rows as independent claims twenty times more information than we have. **This error was made in this work and had to be corrected.**

picture Asking one person twenty questions is not the same as asking twenty people one question. Repeated observations of the same prompt share whatever is peculiar to that prompt, so the twentieth observation adds much less than the first. How much less is exactly what the ICC measures.

- **The model.** $X_{pr} = \mu + a_p + e_{pr}$, with $\mathrm{Var}(a_p) = \sigma_b^2$ ("between prompts") and $\mathrm{Var}(e_{pr}) = \sigma_w^2$ ("within a prompt").
- **The correct variance.** Cluster *means* are independent, so equation (2) applies to them — not to the rows.

$$\mathrm{Var}(\hat{\pi}) = \frac{1}{P}\left(\sigma_b^2 + \frac{\sigma_w^2}{R}\right). \tag{3}$$

Definition 56 (Intraclass correlation)**.** $\rho := \sigma_b^2/(\sigma_b^2 + \sigma_w^2)$: the share of all variation that is *between* prompts. Equivalently, the correlation between two observations drawn from the same prompt.

picture $\rho = 0$: repeats are as good as fresh prompts. $\rho = 1$: all repeats of a prompt are identical and the nineteen extra ones are worthless.

Theorem 57 (Design effect)**.** $\mathrm{DEFF} = \dfrac{\text{equation (3)}}{\sigma^2/(PR)} = 1 + (R-1)\rho.$

Proof. $\dfrac{\frac{1}{P}(\sigma_b^2 + \sigma_w^2/R)}{(\sigma_b^2 + \sigma_w^2)/(PR)} = \dfrac{R\sigma_b^2 + \sigma_w^2}{\sigma_b^2 + \sigma_w^2} = 1 + (R-1)\rho.$ □

one number Measured: $\sigma_b^2 = 0.01361$, $\sigma_w^2 = 0.03677$, so $\rho = 0.2701$ and with $R = 20$, DEFF $= 6.13$. Read: the naive standard error is too small by $\sqrt{6.13} = 2.48$, every naive z is inflated by the same factor, and 19,360 rows carry the information of about 3,150. The honest count of independent units stays 968.

grounded Why the repeats are correlated here, mechanically. Each of the 20 repeats perturbs a *different criterion of the same prompt*. If that prompt's four responses are far apart in score — a wide margin — then *no* single-criterion perturbation flips it, and all twenty repeats return 0 together. If the margin is narrow, many return 1 together. The prompt's margin is a property the repeats **share**, and $\rho = 0.27$ is the size of that sharing.

grounded What it cost. A pairwise-dependence screen reported "89 of 120 pairs survive" computed on the pooled 19,360 rows. Recomputed at the cluster level: 74 of 120. Fifteen apparent findings were the design effect. The derived inflation is $\sqrt{6.13} = 2.48$ and the directly resampled inflation was 2.79; they agree, and the residual is that a correlation is not a simple mean so theorem 57 is only approximate for it.

misread as Row count as sample size. The question is always "how many independent things did I observe", and repeated measurement of the same thing is not independence.

8.15 The bootstrap

problem The formulas above give se for an *average*. For a correlation, a median, a ratio, or a rank statistic there is generally no formula.

picture Replace the algebra with imitation. We cannot rerun the study, but we can rerun a *stand-in* for it: draw a new dataset from the one we have, with replacement, and recompute. The spread of the recomputed values imitates the spread we would have seen across genuinely fresh studies, because the observed sample is the best available stand-in for the population.

Protocol 58 (Cluster bootstrap)**.** Resample P *clusters* with replacement; recompute the statistic; repeat B times; report the 2.5th and 97.5th percentiles.

picture The resampling unit must be the cluster. Resampling rows would build datasets whose internal dependence differs from the real one, reproducing exactly the understatement of theorem 57. Every interval in this paper resamples prompts.

one number 400 resamples of 968 prompts gave $[+0.2566, +0.3520]$ around the observed +0.3040 in §21 — a quantity for which no closed-form se exists.

misread as Bootstrap intervals as assumption-free. They fail with few clusters, at the edge of a parameter's range, and for statistics that are not smooth in the data. All three are checked in §27.

8.16 Permutation tests

problem Sometimes the question is not "how big" but "is the pairing between these two lists doing any work at all".

picture Destroy the pairing and see if the result survives. Shuffle one list against the other many times; if the real pairing gives something the shuffles rarely give, the pairing carries information. It is a *falsifier*: it can say the pairing matters, never *why*.

one number $B = 200$ shuffles: the smallest attainable p is $1/(B+1) = 0.005$. A reported $p = 0.0000$ from 200 shuffles means $p < 0.005$ and nothing more.

grounded What it answers here. Two operators each flip about 5% of prompts. Do they flip the *same* prompts? Shuffling which prompt each operator's outcome is attached to destroys any such coupling while leaving both flip *rates* untouched — so the shuffled distribution is exactly the world where the two operators are unrelated but equally active. Comparing the real coupling to that distribution isolates the coupling and nothing else.

misread as The shuffle as neutral. Shuffling assumes all orderings are equally likely under the null; if the list has internal structure — clustering, time order — shuffling destroys that too, and the test answers a question nobody asked. In this paper shuffles are applied to cluster-level values for exactly this reason.

8.17 Multiplicity

problem Test 170 things at the usual threshold and about 8 will look real when nothing is.

picture Every test is a lottery ticket for a false positive. Buy 170 tickets and you should expect to win. Either raise the bar per test, or change what you are controlling.

Theorem 59 (Bonferroni)**.** *Testing C hypotheses each at level α/C keeps the probability of* any *false rejection at $\leq \alpha$.*

Proof. $\mathbf{1}\{\bigcup A_i\} \leq \sum_i \mathbf{1}\{A_i\}$ pointwise (lemma 8); take expectations via corollary 38: $\mathbb{P}(\bigcup A_i) \leq C \cdot \alpha/C$. □

one number $C = 171$, $\alpha = 0.05$: per-test level 0.000292, whose two-sided normal quantile is $|z| = 3.62$. This paper uses the stricter $|z| > 3.9$ and always reports tests-run beside tests-surviving.

Remark 60 (Bonferroni versus false discovery rate). Bonferroni controls the chance of *any* false positive — right when one false claim is costly. Benjamini–Hochberg instead controls the expected *share* of false positives among rejections, with threshold qk/C at rank k; at $k = C$ that is q itself, not q/C. Confusing the two demands orders of magnitude more resolution than BH requires.

grounded Why 170 tests is not an unusual number. The operator grid in this work has 19 operators; asking which pairs behave alike is $19 \times 18/2 = 171$ questions, and nobody set out to run 171 tests — they fell out of one table. At the usual 5% threshold about 8.5 would look real with nothing there, which is within a factor of two of the number of *genuine* findings. The correction is not pedantry; without it the table is half noise.

misread as Reporting only the survivors. That is the multiplicity failure with manners: the correction is meaningless unless the denominator is published.

8.18 Measurement error, reliability, attenuation

problem Every quantity here is read by a fallible instrument — a language model judging whether a criterion is satisfied. What does that do to the numbers?

picture Measuring with a blurry ruler. Blur adds spread that has nothing to do with the thing being measured, so the measured quantity varies more than the true one. When you correlate two blurry measurements, the shared signal is unchanged but both spreads are inflated — and correlation is signal divided by spreads. So **noise always pushes a correlation toward zero**, never away. It is a systematic pull, not random error, and more data does not fix it.

Definition 61 (Reliability)**.** rel := $\mathrm{Var}(T)/\mathrm{Var}(X) = \sigma_T^2/(\sigma_T^2 + \sigma_\varepsilon^2)$: the share of observed variation that is signal.

Lemma 62 (The identity that makes this free to compute)**.** *For two independent measurements $X_1 = T + \varepsilon_1$, $X_2 = T + \varepsilon_2$ with equal error variances,* $\mathrm{Corr}(X_1, X_2) =$ rel.

Proof. $\mathrm{Cov}(X_1, X_2) = \mathrm{Var}(T)$, since every cross term involves an independent error; each has variance $\sigma_T^2 + \sigma_\varepsilon^2$. □

picture — **Two independent measurements of the same thing agree exactly as much as one of them is trustworthy.** So running a second judge measures the first judge's reliability at no extra conceptual cost — which is what §15 does.

Theorem 63 (Attenuation)**.** $\mathrm{Corr}(X, Y) = \mathrm{Corr}(T, U)\sqrt{\mathrm{rel}_X \mathrm{rel}_Y}$.

Proof. $\mathrm{Cov}(X, Y) = \mathrm{Cov}(T, U)$; and $\mathrm{sd}(X) = \sigma_T / \sqrt{\mathrm{rel}_X}$ from the definition of reliability. Substitute. □

one number — Measured judge reliability 0.5497, so $\sqrt{0.5497} = 0.741$. Read: a *perfect* true relationship, measured through this judge, reads 0.74. No sample size raises it — the pull is bias, and $\sqrt{n}$ does not touch bias.

grounded — The judge, concretely. Two language models scored the same 59,936 criterion-by-response cells; they correlate at 0.5497. By lemma 62 that *is* one judge's reliability, so by theorem 63 every correlation read through this tensor is multiplied by $\sqrt{0.5497} = 0.741$. A relationship that is genuinely perfect reads 0.74; one that genuinely reads 0.4 was really 0.54. **Every number in this paper that routes through satisfaction inherits this factor**, and the compounded ceiling with the panel's own reliability is 0.720.

grounded — And the use that matters. §19 asks whether the two elicitation routes disagree only because both are noisy. Dividing the observed 0.6265 by $\sqrt{0.8702 \times 0.9408}$ removes exactly the blur and leaves 0.6924. The gap does not close — so the disagreement is between what people write and what they choose, not between two noisy views of one thing. **Without theorem 63 that question has no answer at all.**

misread as — A weak measured correlation as a weak true relationship. It may be a strong relationship seen through a blurry instrument, and theorem 63 is the arithmetic that tells them apart — used in §19 to establish that the two elicitations disagree *beyond* their own blur.

8.19 Spearman–Brown: the reliability of a panel

problem — One judge is unreliable. A panel of many should be better. By how much, exactly?

picture — Averaging m measurements multiplies the signal by m but the noise only by $\sqrt{m}$, because independent noise adds in squares (lemma 45). So signal-to-noise improves with panel size, but with diminishing returns — and the formula says precisely how fast.

Theorem 64. *Doubling the number of components takes reliability r to $2r/(1 + r)$.*

Proof. Summing m components gives signal variance $m^2\sigma_T^2$ and error variance $m\sigma_\varepsilon^2$, so $\mathrm{rel}_m = m\lambda/(m\lambda + 1)$ with $\lambda = \sigma_T^2/\sigma_\varepsilon^2$. At $m = 1$, $r = \lambda/(\lambda + 1)$, so $\lambda = r/(1 - r)$; substitute $m = 2$. □

one number — Split-half $r = 0.8961$ gives full-panel $2(0.8961)/1.8961 = 0.9452$. Read: splitting the annotators of a prompt in half and correlating the two halves measures a *half*-sized panel; theorem 64 corrects it to the panel that was actually used, which is the number that enters theorem 63.

8.20 Two-way clustering

problem — Here observations are grouped two ways at once: an annotator appears on many prompts, and a prompt is rated by many annotators. Correcting for one leaves the other.

picture — Two observations are dependent if they share a prompt *or* an annotator. Add the two corrections and you have double-counted the pairs sharing both, so subtract that once. It is inclusion–exclusion applied to covariance terms.

Theorem 65 (Cameron–Gelbach–Miller)**.** $\hat{V}_{\textit{two-way}} = \hat{V}_{\textit{prompt}} + \hat{V}_{\textit{annotator}} - \hat{V}_{\textit{both}}$.

misread as — A slightly negative estimate as a bug. It signals that one dimension carries almost no dependence, which is information.

9 Every tool, executed by hand

Part 0 derived the machinery. This section runs all of it on five numbers, so that nothing is taken on trust. Take the sample

$$x = (2,\ 4,\ 4,\ 4,\ 6), \qquad n = 5.$$

9.1 The mean

$$\bar{x} = \frac{2 + 4 + 4 + 4 + 6}{5} = \frac{20}{5} = 4.$$

9.2 Deviations, and why the naive spread is useless

$$x_i - \bar{x} \;=\; (-2,\ 0,\ 0,\ 0,\ 2), \qquad \sum_i (x_i - \bar{x}) = 0.$$

Exactly zero, as §0.4 said it must be for *any* dataset. This is why the average deviation cannot be a measure of spread.

9.3 Variance, both ways

Squared deviations: $(4, 0, 0, 0, 4)$, summing to 8.

$$s^2 = \frac{8}{n-1} = \frac{8}{4} = 2, \qquad s = \sqrt{2} = 1.4142.$$

Check with lemma 40. $\mathbb{E}[X^2]$ over the sample is $(4+16+16+16+36)/5 = 88/5 = 17.6$, and $\bar{x}^2 = 16$, so the *population-form* variance is $17.6 - 16 = 1.6 = 8/5$. Multiplying by $n/(n-1) = 5/4$ recovers 2. The two forms agree, and the factor between them is exactly lemma 50.

9.4 Standard error and interval

$$\mathrm{se} = \frac{s}{\sqrt{n}} = \frac{1.4142}{2.2361} = 0.6325.$$

By §8.11 the 95% interval is $4 \pm 1.96(0.6325) = 4 \pm 1.2397 = [2.760,\ 5.240]$.

Read it correctly. Across repetitions of the whole study, 95% of such intervals cover the fixed unknown μ. It does *not* say 95% of future observations land in $[2.760, 5.240]$; three of the five observed values are exactly 4, and a prediction interval would be far wider.

9.5 A z-statistic and its p-value

Testing $H_0 : \mu = 0$: $z = 4/0.6325 = 6.32$. From the normal table, $\mathbb{P}(|Z| \geq 6.32) < 10^{-9}$.

Read it correctly. This is $\mathbb{P}(\text{data this extreme} \mid \mu = 0)$. It is not the probability that $\mu = 0$, and $1 - p$ is not the probability the effect is real.

9.6 Minimum detectable effect, and required n

By theorem 54, MDE $= 2.8016 \times 0.6325 = 1.772$: with five observations of this spread, a true mean below 1.772 would be missed more often than not. By corollary 55, detecting $\delta = 0.5$ needs

$$n = \left(\frac{2.8016 \times 1.4142}{0.5}\right)^2 = (7.923)^2 = 62.8 \;\rightarrow\; 63.$$

Detecting $\delta = 0.25$ needs $4\times$ that: 251. This is the $\sqrt{n}$ economics of §8.8 in one line.

9.7 An indicator, a proportion, and its variance

Let P_i be "$x_i > 3$", giving indicators $(0, 1, 1, 1, 1)$. By equation (1) the proportion is $4/5 = 0.8$; by corollary 41 its variance is $0.8 \times 0.2 = 0.16$, so $\mathrm{se} = \sqrt{0.16/5} = \sqrt{0.032} = 0.1789$.

Why rates near the ends are measured better. $\pi(1-\pi)$ peaks at $\pi = 0.5$ with value 0.25 and falls to 0.09 at $\pi = 0.1$. A 5% flip rate is therefore intrinsically easier to pin down than a 50% one, which is why the intervals in table 16 are narrow despite modest n.

9.8 Covariance and correlation

Let $y = (1, 3, 3, 5, 3)$, so $\bar{y} = 3$ and $y_i - \bar{y} = (-2, 0, 0, 2, 0)$.

$$\mathrm{Cov} = \frac{1}{n-1}\sum_i (x_i - \bar{x})(y_i - \bar{y}) = \frac{(-2)(-2) + 0 + 0 + 0 \cdot 2 + 2 \cdot 0}{4} = \frac{4}{4} = 1.$$

$s_y^2 = (4 + 0 + 0 + 4 + 0)/4 = 2$, so $s_y = 1.4142$, and

$$r = \frac{1}{1.4142 \times 1.4142} = \frac{1}{2} = 0.5.$$

Theorem 46 requires $|r| \leq 1$: satisfied. And $r = 0.5$ is *not* "half of anything"; only $r^2 = 0.25$ has a share reading, and only under a linear model.

9.9 The design effect, by hand

Suppose $P = 3$ clusters of $R = 4$:

$$c_1 = (1, 1, 1, 0), \quad c_2 = (0, 0, 0, 0), \quad c_3 = (1, 0, 1, 1).$$

Cluster means: 0.75, 0, 0.75. Grand mean = 0.5.

$$\sigma_b^2 = \tfrac{1}{2}\left[(0.25)^2 + (0.5)^2 + (0.25)^2\right] = \tfrac{1}{2}(0.375) = 0.1875,$$
$$\sigma_w^2 = \text{mean of within-cluster variances} = \tfrac{1}{3}\left[0.25 + 0 + 0.25\right] = 0.1667,$$

using $s^2 = \frac{1}{R-1}\sum(x - \bar{x})^2$ inside each cluster.

$$\rho = \frac{0.1875}{0.1875 + 0.1667} = 0.529, \qquad \text{DEFF} = 1 + (4 - 1)(0.529) = 2.59.$$

So treating the 12 rows as independent would understate the standard error by $\sqrt{2.59} = 1.61$, and the honest count of independent units is 3, not 12.

9.10 Attenuation, by hand

Suppose a true correlation $\text{Corr}(T, U) = 0.90$, measured through instruments with reliabilities 0.55 and 0.95. By theorem 63

$$\text{Corr}(X, Y) = 0.90 \times \sqrt{0.55 \times 0.95} = 0.90 \times \sqrt{0.5225} = 0.90 \times 0.7228 = 0.6505.$$

A strong relationship reads as a moderate one, and no sample size repairs it: the attenuation is bias, and se shrinks while the centre stays wrong.

9.11 Spearman–Brown, by hand

A half-panel split-half correlation of $r = 0.8961$ gives, by theorem 64,

$$\frac{2(0.8961)}{1 + 0.8961} = \frac{1.7922}{1.8961} = 0.9452,$$

which is the reliability of the *full* panel and the number that enters theorem 63.

9.12 Bonferroni, by hand

With $C = 171$ tests at $\alpha = 0.05$: per-test level $0.05/171 = 0.000292$. The two-sided normal quantile at $1-0.000292/2 = 0.999854$ is $|z| = 3.62$. This paper uses 3.9, which is stricter, and reports the count of tests beside the count of survivors.

10 The object

10.1 What a normative disposition is, formally

Definition 66 (Disposition)**.** Let $\mathcal{X}$ be the set of all possible responses to a prompt – unbounded. A person i's *normative disposition* is a preference structure $\succeq_i$ on $\mathcal{X}$. It is not a number, not a sentence, and not a ranking: it is a relation on an infinite set.

Definition 67 (Elicitation)**.** An *elicitation* observes $\succeq_i$ only on a finite *probe set* $X = \{x_1, \ldots, x_k\} \subset \mathcal{X}$. In the object studied here, $k = 4$.

10.2 The chain

$$\succeq_i \;\to\; N_i \;\to\; \mathsf{A} \;\to\; G \;\to\; C \;\to\; J \;\to\; D \;\to\; Y \tag{4}$$

N_i the elicited individual norm; A the aggregation rule; $G = \mathsf{A}(N_1, \ldots, N_n)$ the collective norm; C the compiled standard; J the executor that reads it; D the decision it induces; Y the behaviour of a system trained under it.

Remark 68 (Three kinds of loss, which must never be merged). 1. **Mathematically forced**: no pipeline can avoid it (theorem 74).

2. **Authorised social choice**: aggregation necessarily discards individual information (theorem 84). This is a *defect* only if the rule A was never declared.
3. **Machine distortion**: an unlicensed change.

The released object declares no A. *Therefore categories 2 and 3 are not separable at this site.* That is a missing column, not a hard measurement problem, and it is the single most consequential fact about the release.

10.3 The tensors, exactly

Definition 69 (The released quantities)**.** For prompt p, criterion index c, response letter $r \in \{A, B, C, D\}$, annotator a:

$S[p, c, r] \in (0, 1)$	satisfaction of criterion c by response r;
$W[p, c, a] \in [-10, 10] \setminus \{0\}$	the weight a gave c;
$V[p, a] \subseteq \{A, B, C, D\}$	the responses a called unacceptable;
$\Pi[p, a] \in$ rankings of $\{A, B, C, D\}$	the ranking a gave.

Four facts about definition 69 are load-bearing and are stated here so that no later section can assume otherwise.

1. S is *not* in the release. It is reconstructed by running a language model as a judge; the value is a sigmoid of a logit difference and has never been calibrated against anything. Every number downstream is a number about that judge as much as about the norm (theorem 63, §15).
2. c is *ragged*: between 4 and 39 criteria per prompt, median 15. Any statistic that averages over criteria therefore weights prompts unequally unless it is explicitly re-weighted.
3. W is *sparse*: 102,147 observed (c, a) pairs. Exactly one rating in the corpus equals 0, so the scale is used as strictly two-sided.
4. V **and** W **do not share an index.** V is indexed by (prompt, annotator); W by (prompt, criterion, annotator). There is no column linking a veto to a criterion. §13 shows what this forbids.

Remark 70 (The index itself is reconstructed). The rubric file carries no prompt identifier. The mapping from criteria to prompts is recovered by matching conversation text, with a fuzzy fallback: 966 exact, 2 fuzzy, and 18 of 986 records **never matched at all**. So 1.8% of the rubric corpus is invisible to every number computed from this release, including every number in this paper.

10.4 What a claim about this object can mean

Definition 71 (Induced score)**.** $Y_p(r) \;:=\; \sum_c \bar{w}_{p,c}\, S[p, c, r]$, where $\bar{w}_{p,c}$ is the mean over annotators of $W[p, c, a]$.

Definition 72 (Individual induced score)**.** $N_{i,p}(r) \;:=\; \sum_{c\,:\,i \text{ rated } c} W[p, c, i]\, S[p, c, r].$

Remark 73 (Why N_i must be centred and normalised, and why that is forced). Only contrasts among the four responses decide anything, so adding a constant to N_i changes nothing; the mean over r is therefore unidentified and must be removed. Furthermore, $\|N_i\|$ grows mechanically with the number of criteria i rated: measured $\mathrm{Corr}(\log \|N_i\|, \log \#\{\text{criteria rated}\}) = +0.842$. Any statistic on unnormalised N_i therefore measures *effort*, not values. Centring and normalising are not stylistic; without them the quantity is a different quantity.

11 The elicitation layer

11.1 The probe bound

Theorem 74 (Probe bound)**.** *Let $u : X \to \mathbb{R}$ be a utility on k probe responses. If decisions depend only on differences of u, the space of decision-distinct utilities has dimension $k-1$. If in addition only the ordering matters (no interpersonal scale), the space is S^{k-2}, of dimension $k-2$.*

Proof. S1 The set of utilities is $\mathbb{R}^k$.

S2 u and $u + c\mathbf{1}$ induce identical differences $u(x_a) - u(x_b)$ for every pair, hence identical decisions. The subspace of constants $\mathbb{R}\mathbf{1}$ has dimension 1, so the quotient $\mathbb{R}^k/\mathbb{R}\mathbf{1}$ has dimension $k-1$. Concretely, the centring map $C = I - \frac{1}{k}\mathbf{1}\mathbf{1}^\top$ is a projection with $\mathrm{rank}(C) = k-1$; for $k = 4$ this rank is verified numerically as 3.

S3 If u and λu for $\lambda > 0$ are also equivalent, we quotient the $(k-1)$-dimensional centred space by the positive rays, giving the sphere S^{k-2}, of dimension $k-2$.

S4 For $k = 4$: dimension 3 after centring, and S^2 – two free parameters – after scale.

□

Theory claim. However many criteria a person writes and however many weights they attach, the elicitation returns **a point on a 2-sphere**: two free numbers per person per prompt. This is arithmetic, not a finding, and it is the ceiling on everything downstream.

Corollary 75 (Three consequences)**.** *1. The numerator of "how much normative information did the pipeline destroy" is bounded by two parameters. Most of what a criterion says never entered the measurable object at all.*

2. Any measure of loss based on rank *or* dimension *must read a constant. An earlier round of this work built such a measure; it returned the value* 3.00 *on every input including its own null. That was not a bug – it was theorem 74.*

3. Loss must be measured as angle *and* influence*. You cannot lose a dimension you never had; you can only be rotated away from what people wanted.*

11.2 Is the weight scale ordinal or interval? A test, and an answer

Nothing in the elicitation establishes that the distance from +8 to +10 means the same as from +2 to +4. If the scale is merely *ordinal* – only the ordering of criteria within an annotator is meaningful – then any strictly increasing transformation of the weights must leave every decision unchanged.

Protocol 76 (Monotone gauge test)**.** Apply φ with φ strictly increasing and odd, recompute Y_p from definition 71 with $\varphi(\bar{w})$ in place of $\bar{w}$, and record whether $\arg\max_r Y_p(r)$ moves.

transform $\varphi(w)$	winner changes
identity (control)	0.0%
$\mathrm{sign}(w)\lvert w\rvert^{1/2}$	4.4%
$\mathrm{sign}(w)\log(1+\lvert w\rvert)$	4.6%
$\mathrm{sign}(w)\lvert w\rvert^{2}$	7.3%
$\mathrm{sign}(w)$ alone	**11.7%**

Table 7: Every transform preserves each annotator's ordering of criteria exactly. If the scale were ordinal, all rows would read 0.0%. $n = 968$ prompts.

Boundary. The scale is *used* as an interval scale. 7.3% of decisions change under a transform that preserves every stated preference ordering, and 11.7% change when magnitude is discarded entirely. Every result in this literature that sums weighted criteria depends on an interval assumption the elicitation never established. The last row is the honest headline: roughly one decision in eight is carried by magnitude rather than direction.

11.3 What the elicitation captured: consensus, not the person

quantity (14,764 person–prompt pairs)	value
$\cos(N_i,$ person i's own ranking)	+0.3928
$\cos(N_i,$ a *different* person's ranking)	+0.3097
difference	+0.0831 (se 0.0067, $z = +12.3$)

Table 8: Both vectors centred and normalised per remark 73.

Two readings, and the second is larger than the first.

- **Relative.** The entire individual signal is 0.083 of cosine. A person's written criteria predict a *stranger's* ranking almost as well as their own.
- **Absolute.** 0.393 means a person's own stated criteria recover only about 39% of the direction of their own choice.

Theory claim. Before the pipeline does anything, roughly 61% of the relation between what a person *writes* and what they *choose* is already absent. No downstream step can destroy individual normative content that the elicitation never captured; "how much of a person survives compilation" is bounded above by how much of them was ever written down.

11.4 The normative payload actually collected

field, by lexical marker	criteria	share
exception / carve-out (*unless, except, other than*)	123	0.8%
conditional scope (*if, when, where, in cases of*)	1,724	11.3%
priority / conflict (*before, over, takes precedence*)	440	2.9%
uncertainty (*may, might, unclear, depends*)	478	3.1%
non-compensable (*never, must not, regardless*)	121	0.8%
none of the five	**12,547**	**82.3%**

Table 9: All 15,248 criteria. Median length 94 characters.

A minimal normative payload has been proposed with thirteen fields: content, scope, polarity, exceptions, priority, type, compensability, authority, judgment type, uncertainty, witnesses, constituency, meta-rules. This release populates **three**: content, polarity (the sign of a weight), magnitude.

Corollary 77 (A question that is empty here)**.** *"Did the compilation destroy the exception clause?" cannot be studied at this site:* 99.2% *of criteria contain no exception. A pipeline cannot destroy structure the elicitation never collected. This is a statement about the release, not about pipelines.*

12 The aggregation layer

12.1 Social choice, from zero

Given n people each with a ranking of k alternatives, a *social choice rule* outputs one alternative. Five rules appear below; each is defined completely.

Definition 78 (Rules)**.** Let $\pi_i(x)$ be the Borda points person i gives x (i.e. $k-1$ for their first choice down to 0 for last, ties sharing the average).

- **Plurality**: pick the x that is ranked first by the most people.
- **Borda**: pick $\arg\max_x \frac{1}{n}\sum_i \pi_i(x)$.
- **Median**: pick $\arg\max_x \operatorname{median}_i \pi_i(x)$.
- **Maximin**: pick $\arg\max_x \min_i \pi_i(x)$ – the alternative whose *worst* treatment by any individual is least bad.
- **Condorcet winner**: the x that beats every other alternative in pairwise majority comparison. It need not exist.

Definition 79 (Condorcet cycle)**.** A majority cycle is a triple with x beating y, y beating z, and z beating x. Majority preference is then not transitive and no Condorcet winner exists.

12.2 Measured social choice structure

quantity (968 prompts, ≥ 3 rankings each)	value
a Condorcet winner exists	72.3%
a majority *cycle* occurs	**0.0**%
neither (majority ties)	27.7%

Table 10: The classic voting pathology does not occur in this corpus.

agreement with the Condorcet winner, where one exists ($n = 700$)	
Borda	98.4%
Plurality	98.3%
Median	97.4%
the weighted-criteria score of definition 71	**68.4**%
Maximin	55.9%

Theory claim. Every standard *ranking-based* rule agrees with the Condorcet winner to 97–98%. The weighted-criteria score agrees only 68.4% of the time. The disagreement is therefore **not** between aggregation rules; it is between *scoring people's criteria* and *counting people's rankings* – two elicitations from the same people whose outputs differ on nearly a third of prompts.

12.3 Arrow's independence condition, measured

Definition 80 (IIA)**.** A rule satisfies *independence of irrelevant alternatives* if the social ordering of x and y depends only on individual orderings of x and y, not on any third alternative.

Measured for Borda: over 11,616 tests (each prompt × each dropped response × each remaining pair), the relative order of two responses reverses when an irrelevant third is removed in 1,553 cases: **13.4**%.

Corollary 81 (The candidate set is part of the decision)**.** 13.4% *of pairwise relations flip when the option set changes. Any normative content measured on one set of four responses is, to that extent, a property of the set. This is the quantitative content of "candidate-set overfitting".*

12.4 Blackwell: what aggregation necessarily costs

Definition 82 (Garbling)**.** G is a *garbling* of Z if there is a Markov kernel K with $G \sim K(\cdot \mid Z)$: G is generated from Z by a channel that adds no new information about the state.

Lemma 83 (Aggregation is a garbling)**.** *If $G = \mathsf{A}(Z)$ is a deterministic function of Z, then G is a garbling of Z, via the degenerate kernel $K(\cdot \mid z) = \delta_{\mathsf{A}(z)}$.*

Theorem 84 (Blackwell, invoked)**.** *If G is a garbling of Z then for* every *decision problem d with any loss function, the attainable Bayes risk satisfies $R_G(d) \geq R_Z(d)$.*

Corollary 85. *Aggregation never adds information and weakly loses on every decision problem. The question is never* whether, *only* how much *– and "how much" is meaningless until a decision family is named.*

Definition 86 (Deficiency)**.** $\delta_{\mathcal{D}}(Z, G) := \sup_{d \in \mathcal{D}} \big[R_G(d) - R_Z(d)\big]$.

Computing it here. Take $\mathcal{D}$ to be the single problem "choose one of the four responses". Normalise each person's induced score to $[0, 1]$ by min–max within that person and prompt, so that scores are interpersonally comparable by construction rather than by assumption. Then:

$$\begin{aligned} \text{oracle (each person receives their own top response)} &= 1.0000 \\ \text{the single collective choice} &= 0.8702 \\ \textbf{deficiency} &= \mathbf{0.1298} \end{aligned}$$

with quartiles 0.0445/0.1076/0.1924 across prompts.

Theory claim. Thirteen points of normalised utility is what *disagreement itself* costs. It is a social choice, not a pipeline defect, and no compilation recovers it. It also fixes the denominator: "what fraction was preserved" should be divided by 0.8702, not by 1.

12.5 Sen's liberal paradox, instantiated

prompts carrying a partial veto	161
the weighted-criteria winner *is* a vetoed response	42 = **26.1**%

One time in four, a rule that maximises aggregate satisfaction selects an option someone declared unacceptable. This is not a defect of the rule; it is what a utilitarian aggregator does. It is the empirical half of §13.

12.6 Which rule is the larger lever, corrected

An earlier version of this analysis applied "median" and "maximin" across *criteria*. A social choice rule aggregates across *people*; the two coincide only if each criterion belongs to exactly one person, which is false here. Corrected:

rule, versus the mean over people	winner changes	(wrong index: over criteria)
maximin	49.4%	63.9%
median	14.8%	34.3%
Borda	9.6%	–
trimmed	**3.6**%	–
any intervention on a single criterion	2.1–11.6%	

Remark 87 (And maximin needs a control). A minimum over a set is unstable by construction. Control: same people, same per-person mean and standard deviation, response pattern destroyed. That sham disagrees with the mean rule on 61.4% of prompts – *more* than the real data's 49.4%. So maximin's high rate is the instability of the min operator, and the actual structure of human disagreement *reduces* it by 12 points. The claim "minority-sensitive rules change half the outcomes" does not survive its own control. What survives is that *trimmed* aggregation, at 3.6%, moves less than several criterion-level interventions: trimming the extremes barely matters, so the disagreement that matters lives in the tails.

13 The type layer: why a veto is not a large negative weight

13.1 The index mismatch

From definition 69: V is indexed (p, a) and W is indexed (p, c, a). An operator "represent this veto as a weight" must choose *which criterion* carries it. No such column exists.

Proposition 88. *"Coerce a veto into a criterion weight" is not a function of the released data.*

This is a schema fact, not a measurement difficulty, and no amount of data at this site repairs it.

13.2 The well-posed question, and its answer

Definition 89 (Two rules)**.**

$$\text{constraint:} \quad a^\star = \arg\max_{a \notin V} Y(a), \tag{5}$$

$$\text{penalty:} \quad a^\star_M = \arg\max_{a} \left[Y(a) - M \cdot \mathbf{1}\{a \in V\} \right]. \tag{6}$$

Theorem 90 (Exact condition for a penalty to reproduce a veto)**.** *Let $\Delta := \max_{a \in V} Y(a) - \max_{a \notin V} Y(a)$. Then $a^\star_M = a^\star$ for every realisation of the scores if and only if $M > \Delta$.*

Proof. *(⇐)* Suppose $M > \Delta$. For any $a \in V$, $Y(a) - M \le \max_{a' \in V} Y(a') - M < \max_{a' \in V} Y(a') - \Delta = \max_{a' \notin V} Y(a')$. So every vetoed alternative is strictly below the best allowed one after the penalty, and the arg max lies in the allowed set, where the penalty is zero and the ordering is unchanged. Hence $a^\star_M = a^\star$.

(⇒) Suppose $M \le \Delta$. Let a_V attain $\max_{a \in V} Y(a)$. Then $Y(a_V) - M \ge Y(a_V) - \Delta = \max_{a \notin V} Y(a)$, so a_V is weakly preferred by the penalty rule to every allowed alternative, and the penalty rule may select a vetoed response. Hence agreement fails. □

measured quantity	value
prompts with a partial veto	161
the veto is *binding* ($\Delta > 0$)	42 = 26.1%
required M: median	3.64
required M: 90th percentile	9.92
required M: **maximum**	**26.54**
contribution of one criterion at the scale maximum	≤ 10
required M in units of max-weight criteria: median / max	0.36 / **2.65**

Theorem 91 (No finite weight encodes a veto)**.** *Δ is a function of the* candidate set*, not of the norm. Since a vetoed response can be arbitrarily better than every allowed one on the remaining criteria,* $\sup_X \Delta(X) = \infty$*. Therefore no finite M satisfies theorem 90 uniformly over candidate sets.*

Proof. Fix a person, a veto set V, and a criterion c with weight $w > 0$. Construct a candidate $x^* \in V$ with $S[\cdot, c, x^*] \to 1$ and all allowed candidates with $S[\cdot, c, \cdot] \to 0$; then $\Delta \to w \cdot \sum$ over however many such criteria are present, which is unbounded as the number of agreeing criteria grows. Since theorem 90 requires $M > \Delta$ for agreement on that set, no fixed finite M suffices for all sets. □

Theory claim. A veto is a property of the *person*. The penalty M that reproduces it is a property of the person **and the four responses that happened to be shown**. On just 968 four-item candidate sets the required penalty already reaches 2.65 times the entire elicitation scale. Type is not compressible to magnitude, and this is a theorem with its counterexample present in the data – it required no instrument at all.

14 The compilation layer

The released standard ships in two forms: the full weighted rubric and a compiled "core" whose items carry no weights, every one rewritten to positive polarity.

- The two select *different* winners on **29.4**% of prompts.
- Compilation performs three operations at once – selection, rewriting, and weight discarding – and the release ships no lineage, so the three are **not separable**. This is the missing column that blocks the most questions here.

14.1 How many bits of the rubric are decision-relevant?

Protocol 92 (Rate–distortion)**.** Quantise every criterion weight to b bits uniformly across the observed range within each prompt, recompute definition 71, and record whether the winner is preserved.

bits per criterion weight	decision preserved
1 (sign only)	**89.8**%
2	95.0%
3	98.1%
4	98.6%
8	99.8%

Remark 93 (Two independent computations agree). section 14.1 says one bit preserves 89.8%, i.e. changes 10.2%. table 7 says discarding magnitude changes 11.7%. These were computed by different routes and agree to about a point. The 21 levels of the $-10..+10$ scale carry roughly three bits of decision-relevant content, and the *first* bit carries nine tenths of it.

Corollary 94 (A design consequence)**.** *The elicitation is over-precise in magnitude and empty in type (table 9). The next collection should not buy finer weights; it should buy fields.*

15 The instrument layer

15.1 The judge as a noisy channel

S is produced by a language model. Two independent judges give two measurements of the same underlying satisfaction, so lemma 62 applies directly.

59,936 satisfaction cells	value
$\mathrm{Corr}(\mathrm{judge}_1, \mathrm{judge}_2)$	0.5497
$\Rightarrow$ reliability of one judge (lemma 62)	0.5497
$\Rightarrow$ attenuation factor $\sqrt{\mathrm{rel}}$ (theorem 63)	0.741

Corollary 95 (A ceiling on every number in this literature)**.** *A true correlation of* 1.0 *measured through this tensor reads at most* 0.741*. This is bias, not noise: more prompts do not remove it.*

15.2 Panel reliability

Split-half over annotators on 964 prompts with $\geq$ 6 annotators: $r = 0.8961$; by theorem 64, full-panel reliability $2r/(1+r) = 0.9452$, so the panel score alone caps correlations at $\sqrt{0.9452} = 0.9722$.

Corollary 96 (Compounded ceiling)**.** *A correlation measured through both the panel and the judge is capped at* $0.9722 \times 0.741 = 0.720$.

15.3 The gauge test

Definition 97 (Gauge transformation)**.** A transformation that must leave the *property* unchanged. Here: change the reader of the norm while holding the norm fixed. If the *measurement* is invariant but the property is not, the measurement is blind; if the property is invariant but the measurement is not, the measurement is about the reader.

	winner differs from baseline	n
judge B (different family)	42.4%	968
judge C (different scale)	39.7%	968
same judge, responses reordered	**8.0**%	300

Table 11: (a) The object is *not* invariant.

Spearman ρ of the operator ordering, against baseline	judge B	judge C	reordered
	+0.982	+1.000	+0.963

Table 12: (b) The measurement *is* invariant. Pre-registered kill at $\rho < 0.8$; it does not fire on any of the three.

Theory claim. Comparative statements are about the norm. Absolute statements are about the reader. "Operator X moves the decision more than Y" survives three instruments and a position swap. "The standard prefers response X" does not: roughly 40% of it is the reader and 8 points are nothing but the order the responses were printed in.

Remark 98 (The scale reference, with its caveat). Reordering the page changes the winner more often than deleting a criterion does (8.0% versus 5.1%). The caveat is load-bearing: the reordering comparison uses a *separate* judge run and so carries judge stochasticity, whereas the operator comparison holds the tensor fixed and cancels it. So 8.0% is not a strict noise floor for the operators; it is the correct *scale reference* for any claim of the form "this changes the decision".

16 The whole chain on numbers you can check

Everything in §11–§21 is now executed on a three-person, four-response, three-criterion example. Every number below can be verified with a pencil.

16.1 The setup

Three annotators $\{1, 2, 3\}$, four responses $\{A, B, C, D\}$, three criteria $\{c_1, c_2, c_3\}$. The judge returns:

$$S = \begin{pmatrix} & A & B & C & D \\ c_1 & 0.9 & 0.8 & 0.2 & 0.1 \\ c_2 & 0.2 & 0.7 & 0.9 & 0.3 \\ c_3 & 0.5 & 0.4 & 0.4 & 0.9 \end{pmatrix}$$

The weights, one row per annotator (a blank means that annotator did not rate that criterion):

$$W = \begin{pmatrix} & c_1 & c_2 & c_3 \\ 1 & +8 & +2 & -1 \\ 2 & +6 & +4 & \cdot \\ 3 & -2 & +9 & +3 \end{pmatrix}$$

Annotator 3 additionally declares $V_3 = \{A\}$ unacceptable.

16.2 Individual norms N_i (definition 72)

$$
\begin{aligned}
N_1 &= 8(0.9, 0.8, 0.2, 0.1) + 2(0.2, 0.7, 0.9, 0.3) - 1(0.5, 0.4, 0.4, 0.9)\\
&= (7.2, 6.4, 1.6, 0.8) + (0.4, 1.4, 1.8, 0.6) - (0.5, 0.4, 0.4, 0.9)\\
&= (7.1,\ 7.4,\ 3.0,\ 0.5)\\
N_2 &= 6(0.9, 0.8, 0.2, 0.1) + 4(0.2, 0.7, 0.9, 0.3) = (5.4, 4.8, 1.2, 0.6) + (0.8, 2.8, 3.6, 1.2) = (6.2,\ 7.6,\ 4.8,\ 1.8)\\
N_3 &= -2(0.9, 0.8, 0.2, 0.1) + 9(0.2, 0.7, 0.9, 0.3) + 3(0.5, 0.4, 0.4, 0.9)\\
&= (-1.8, -1.6, -0.4, -0.2) + (1.8, 6.3, 8.1, 2.7) + (1.5, 1.2, 1.2, 2.7)\\
&= (1.5,\ 5.9,\ 8.9,\ 5.2)
\end{aligned}
$$

Individual top choices: $N_1 \to B$, $N_2 \to B$, $N_3 \to C$.

16.3 Centring and normalising (remark 73)

N_1 has mean $(7.1+7.4+3.0+0.5)/4 = 4.5$, so centred it is $(2.6, 2.9, -1.5, -4.0)$ with norm $\sqrt{6.76 + 8.41 + 2.25 + 16} = \sqrt{33.42} = 5.781$, giving

$$\hat{N}_1 = (0.4497,\ 0.5016,\ -0.2595,\ -0.6919).$$

Note $\|N_1\|$ before centring was dominated by the common level, which decides nothing – exactly the point of remark 73. Annotator 2 rated only two criteria and would have a systematically smaller raw norm; after normalisation the two are comparable.

16.4 Aggregation, three ways

Mean over people. $\bar{N} = \frac{1}{3}(N_1 + N_2 + N_3) = \frac{1}{3}(14.8,\ 20.9,\ 16.7,\ 7.5) = (4.93,\ 6.97,\ 5.57,\ 2.50)$, so the winner is B.

Median over people. Componentwise medians of the three vectors: $\mathrm{med}(7.1, 6.2, 1.5) = 6.2$; $\mathrm{med}(7.4, 7.6, 5.9) = 7.4$; $\mathrm{med}(3.0, 4.8, 8.9) = 4.8$; $\mathrm{med}(0.5, 1.8, 5.2) = 1.8$. Winner B.

Maximin over people. Componentwise minima: $(1.5,\ 5.9,\ 3.0,\ 0.5)$. Winner B. Note the minimum is attained by a *different* person in each column – annotator 3 for A, annotator 3 for B, annotator 1 for C – which is why min is unstable: a small change to any one person can change which person defines the value.

The weighted-criteria route (definition 71). Mean weights are $\bar{w} = (\frac{8+6-2}{3},\ \frac{2+4+9}{3},\ \frac{-1+3}{2}) = (4,\ 5,\ 1)$, so

$$Y = 4(0.9, 0.8, 0.2, 0.1) + 5(0.2, 0.7, 0.9, 0.3) + 1(0.5, 0.4, 0.4, 0.9) = (5.1,\ 7.1,\ 5.7,\ 2.8),$$

winner B. Here the two routes agree; §12 reports that on the real corpus they disagree 31.6% of the time.

16.5 Condorcet, by hand

Individual rankings from the N_i: person 1: $B > A > C > D$; person 2: $B > A > C > D$; person 3: $C > B > D > A$. Pairwise majorities: B beats A (3–0), B beats C (2–1), B beats D (3–0). B is the Condorcet winner, and it coincides with all four rules above.

Constructing a cycle, to show it is possible. With rankings $A > B > C$, $B > C > A$, $C > A > B$: A beats B (2–1), B beats C (2–1), C beats A (2–1). No Condorcet winner exists. The measured corpus contains *zero* such cycles (table 10), which is a genuine negative result and not an assumption.

16.6 IIA violation, by hand

Take Borda points on four alternatives $(3, 2, 1, 0)$. With the three rankings above (person 3 ranking $C > B > D > A$):

$$A : 2 + 2 + 0 = 4, \quad B : 3 + 3 + 2 = 8, \quad C : 1 + 1 + 3 = 5, \quad D : 0 + 0 + 1 = 1.$$

So $C > A$ socially, by 5 to 4. Now delete D and recompute on three alternatives $(2, 1, 0)$: person 1: $B > A > C$, person 2: same, person 3: $C > B > A$.

$$A : 1 + 1 + 0 = 2, \quad B : 2 + 2 + 1 = 5, \quad C : 0 + 0 + 2 = 2.$$

Now C and A are *tied*, where before C led. Deleting an alternative nobody ranked first changed the relation between two others. That is an IIA violation, and it occurs on 13.4% of tests in the real corpus (corollary 81).

16.7 The veto condition, by hand (theorem 90)

Annotator 3 vetoes A. From $Y = (5.1, 7.1, 5.7, 2.8)$:

$$\Delta = \max_{a \in V} Y(a) - \max_{a \notin V} Y(a) = Y(A) - \max(7.1, 5.7, 2.8) = 5.1 - 7.1 = -2.0 < 0.$$

The veto is *not binding*: A was not going to win anyway, so any $M \geq 0$ reproduces the constraint.

Now change one number: let $S[c_1, A] = 0.9 \to 1.0$ and $\bar{w}_1 = 4 \to 9$. Then $Y(A) = 9(1.0) + 5(0.2) + 1(0.5) = 10.5$ and $Y(B) = 9(0.8)+5(0.7)+1(0.4) = 11.1$, $Y(C) = 9(0.2)+5(0.9)+1(0.4) = 6.7$, $Y(D) = 9(0.1)+5(0.3)+1(0.9) = 3.3$. Still $\Delta = 10.5 - 11.1 < 0$. Push once more, $\bar{w}_1 = 20$: $Y(A) = 21.5$, $Y(B) = 21.9$ – the gap closes but never inverts, because B also satisfies c_1 highly. To make Δ large one needs a vetoed response that is *uniquely* good on a heavily weighted criterion, and there is no bound on how good: that is precisely the construction in theorem 91, and on the real corpus the realised maximum is $\Delta = 26.54$ against a per-criterion ceiling of 10.

16.8 Regret versus flip rate, by hand (lemma 115)

Take $Y = (5.1, 7.1, 5.7, 2.8)$, so $a^\star = B$, $Y(a^\star) = 7.1$, $\min_a Y = 2.8$, and the denominator of definition 113 is $7.1 - 2.8 = 4.3$.

- A perturbation that shifts the winner to C: $\mathrm{Reg} = (7.1 - 5.7)/4.3 = 0.326$.
- A perturbation that shifts the winner to A: $\mathrm{Reg} = (7.1 - 5.1)/4.3 = 0.465$.
- A perturbation that shifts the winner to D: $\mathrm{Reg} = (7.1 - 2.8)/4.3 = 1.000$.

All three are "a flip". The flip rate assigns them the same value; regret separates them by more than a factor of three. Lemma 115 says $\mathbb{E}[\mathrm{Reg}] = \pi \cdot \mathbb{E}[\mathrm{Reg} \mid \mathrm{flip}]$ exactly, and it is the second factor – 0.326 versus 1.000 here – that the flip rate discards.

16.9 The margin cap, by hand (proposition 118)

The margin is $m = 7.1 - 5.7 = 1.4$, and the score range is $7.1 - 2.8 = 4.3$, so the relative margin is 0.326 – almost exactly the corpus median of 0.319. A perturbation δ can only flip the winner if it moves the top-two gap past zero, so by proposition 118 it needs $\|\delta\|_\infty > m/2 = 0.7$, i.e. about 16% of the range. Perturbations smaller than that cannot flip this prompt whatever they destroy normatively – which is the cap, made concrete.

16.10 Shapley influence, by hand, and it exhibits C1 directly

Three annotators, so $3! = 6$ orderings. The collective winner is B. The characteristic function $v(S)$ asks whether the coalition S, summing its own N_i, also picks B. Enumerating all eight subsets:

$$v(\varnothing) = 0, \quad v(\{1\}) = 1, \quad v(\{2\}) = 1, \quad v(\{3\}) = 0,$$

$$v(\{1,2\}) = 1, \quad v(\{1,3\}) = 1, \quad v(\{2,3\}) = \mathbf{0}, \quad v(\{1,2,3\}) = 1.$$

The one to check by hand is $v(\{2,3\})$: $N_2 + N_3 = (7.7,\ 13.5,\ 13.7,\ 7.0)$, whose maximum is at C, not B.

Marginal contributions, all six orderings.

ordering	ϕ_1 term	ϕ_2 term	ϕ_3 term
1, 2, 3	$v(1) - v(\varnothing) = 1$	$v(12) - v(1) = 0$	$v(123) - v(12) = 0$
1, 3, 2	1	$v(123) - v(13) = 0$	$v(13) - v(1) = 0$
2, 1, 3	$v(12) - v(2) = 0$	$v(2) - v(\varnothing) = 1$	$v(123) - v(12) = 0$
2, 3, 1	$v(123) - v(23) = 1$	1	$v(23) - v(2) = -1$
3, 1, 2	$v(13) - v(3) = 1$	$v(123) - v(13) = 0$	$v(3) - v(\varnothing) = 0$
3, 2, 1	$v(123) - v(23) = 1$	$v(23) - v(3) = 0$	0
total	5	2	-1

$$\phi_1 = \tfrac{5}{6} = 0.8333, \qquad \phi_2 = \tfrac{2}{6} = 0.3333, \qquad \phi_3 = -\tfrac{1}{6} = -0.1667.$$

Two checks. Efficiency (axiom E of §17): $\frac{5}{6} + \frac{2}{6} - \frac{1}{6} = \frac{6}{6} = 1 = v(A) - v(\varnothing)$. And a *negative* value is legitimate: annotator 3 actively pushes the coalition away from the collective winner, so their average marginal contribution to selecting B is below zero. Alignment, being a cosine of a person against a sum that includes them, cannot go negative in the same situation and therefore cannot express this at all.

Now the alignment of each, against the collective. The collective centred vector is $\bar{N}-4.9917 = (-0.06,\ 1.97,\ 0.57,\ -2.49)$ (up to rounding), and centring each N_i gives cosines

$$\cos(N_1, G) = 0.7858, \qquad \cos(N_2, G) = \mathbf{0.9313}, \qquad \cos(N_3, G) = 0.2189.$$

And there is the point of §21, in three people.

annotator	alignment	influence ϕ_i	
1	0.7858	**+0.8333**	highest influence, *second* alignment
2	**0.9313**	+0.3333	highest alignment, *less than half* the influence
3	0.2189	−0.1667	lowest alignment, and *negative* influence

The rank orders disagree at the top: annotator 2 agrees with the outcome most and moves it least of the two contributors. The reason is visible in the enumeration – annotator 2 alone already selects B, so in the orderings where they arrive after annotator 1 they add nothing. *Agreeing with an outcome that would have happened anyway is not influence over it.* That is the entire content of C1, and on the real corpus it appears as a correlation of +0.304 with a profile that flattens above moderate agreement (table 17).

17 The three classical theorems, proved rather than cited

A reader with no background cannot check an invoked theorem. The three that carry weight in this paper are therefore proved or, where the full proof is long, reduced to a checkable core.

17.1 Blackwell: why a deterministic aggregation can only lose

Theorem 99 (Garbling implies weakly higher risk)**.** *Let Z be an observation and $G = K(Z)$ for a Markov kernel K. For any decision problem with action set $\mathcal{A}$, state θ, prior p and loss L, the minimum attainable expected loss using G is at least that using Z.*

Proof. S1 A decision rule based on G is a map $d_G : G \mapsto \mathcal{A}$.

S2 Composing with the kernel, $d_G \circ K$ is a (possibly randomised) decision rule based on Z: given z, draw $g \sim K(\cdot \mid z)$ and play $d_G(g)$.

S3 By construction the two procedures induce the same joint distribution over (θ, action), hence the same expected loss.

S4 Therefore every risk attainable with G is attainable with Z; the set of Z-attainable risks contains the set of G-attainable risks; and a minimum over a superset is no larger. Hence $R_Z \le R_G$.

□

Corollary 100. *With $K(\cdot \mid z) = \delta_{\mathsf{A}(z)}$ (a deterministic aggregation), the theorem applies verbatim: aggregation cannot help on any decision problem, and the deficiency $\delta_{\mathcal{D}}$ is well defined and non-negative.*

Remark 101 (What is *not* proved here). The converse – that if $R_Z \le R_G$ for every decision problem then G must be a garbling of Z – is Blackwell's substantive theorem and is not needed. Only the easy direction is used, and it is the direction that licenses "aggregation never adds information".

17.2 Arrow: the statement, and why the count 13.4% is the right thing to measure

Theorem 102 (Arrow, stated)**.** *For $k \ge 3$ alternatives, no social welfare function on the full domain of individual orderings simultaneously satisfies: (a) unrestricted domain, (b) unanimity (if everyone prefers x to y then society does), (c) independence of irrelevant alternatives, and (d) non-dictatorship.*

Why this paper measures rather than invokes it. Arrow says the four conditions are jointly unsatisfiable; it does not say *how badly* a given rule fails a given condition on a given corpus. Borda is a perfectly reasonable rule that fails only IIA, and the useful empirical question is: on this data, how often does the failure actually bite? The answer is 13.4% of pairwise relations (corollary 81), and *that* is the number a designer needs, not the impossibility.

The core of the failure, in one line. Borda scores depend on *how many* alternatives a given one is above, so inserting or removing a third alternative changes the points of the other two by different amounts whenever individuals rank the newcomer differently. The worked example in §16 shows the mechanism on three rankings.

17.3 Shapley: the axioms, and why they force the formula

Definition 103 (Axioms)**.** A value ϕ assigning a number to each player of a cooperative game v satisfies:

(E) **Efficiency**: $\sum_i \phi_i = v(A) - v(\varnothing)$.

(S) **Symmetry**: if i and j contribute identically to every coalition, $\phi_i = \phi_j$.

(N) **Null player**: if i adds nothing to any coalition, $\phi_i = 0$.

(L) **Linearity**: $\phi(v + w) = \phi(v) + \phi(w)$.

Theorem 104 (Uniqueness, core of the argument)**.** *The Shapley value is the unique ϕ satisfying (E), (S), (N), (L).*

Reduction to a checkable core. S1 The *unanimity games* u_T, defined by $u_T(S) = 1$ if $T \subseteq S$ and 0 otherwise, form a basis of the vector space of all games on A: every v has a unique expansion $v = \sum_T \lambda_T u_T$.

S2 On u_T, axioms (E), (S), (N) force the value uniquely: players outside T are null, so get 0 by (N); players inside T are symmetric, so share equally by (S); and the total is 1 by (E). Hence $\phi_i(u_T) = 1/|T|$ for $i \in T$ and 0 otherwise.

S3 By (L), $\phi(v) = \sum_T \lambda_T \phi(u_T)$, which is now determined. Rearranging that sum into a sum over coalitions yields the permutation-weighted formula of §21.

□

Remark 105 (Why this matters for the claim being made). The Shapley value is not chosen here for convenience. It is the *only* assignment of credit satisfying four properties one would insist on for a claim about influence – in particular (N), which guarantees that an annotator who never changes any outcome receives exactly zero. A correlational measure of alignment has no such guarantee, and table 17 shows the gap is real: the highest-alignment quintile does not have the highest influence.

18 Measurement contracts and the Evidence Firewall

18.1 The Evidence Firewall

Firewall principle. A measurement is evidence about the Machine; it is not the Machine. An interpretability output is licensed by a declared measurement method and experimental contract. It is not, by itself, a computational variable, native variable, causal mechanism, or privileged coordinate system of the model.

The firewall is not skepticism about measurement. It is a type discipline: a probe result, sparse code, attribution score, interchange effect, reconstruction, and transplant answer different questions. Promotion to a stronger claim requires the missing experiment rather than a change in vocabulary.

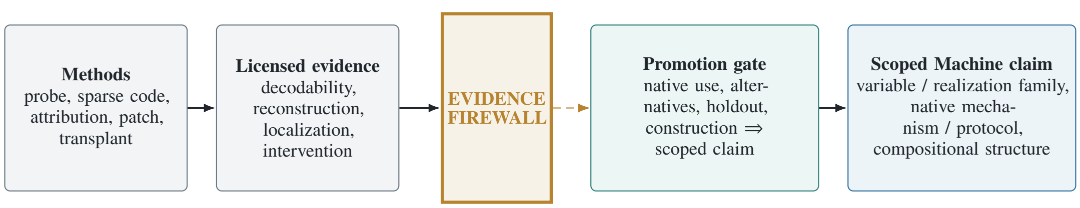

Figure 1: The Evidence Firewall. A measurement method generates evidence; its declared contract states what that evidence permits, and the firewall blocks every stronger default inference. The dashed crossing is licensed only by the named promotion tests; the final solid arrow records the resulting scoped claim. The diagram is an evidence-licensing order, not a model-execution graph.

Table 14: Default licensed conclusions. A method may support a stronger conclusion only by supplying the missing evidence.

Evidence	Default license	Does not by itself license
Held-out probe	Decodability in a specified class	Native use, causality, natural coordinates
Sparse dictionary	Sparse reconstruction in a chosen basis	Causal-variable, Jacobian, or pathway sparsity
Attribution/mask	A declared localization estimand	Necessity, sufficiency, or interaction completeness
Natural-donor interchange	Counterfactual use within support	Global identity or unique mechanism
Ablation/rescue	Scoped necessity/sufficiency	Portability or compositional closure
Reconstruction / transplant	Constructive actionability	Metaphysical uniqueness
Withheld composition	Typed compositional prediction	Higher context laws outside the test scope

Table 13: The CTM evidence matrix. Color or architectural location never assigns a role; each role requires a distinct counterfactual test. Coordination claims require factorial or coalition experiments across roles.

Role	Candidate sensors	Decisive intervention	Forbidden inference
Compute	Local input–output map, Jacobian/finite differences, operation probes, component attribution	Interchange inputs while holding the physical memory realization and route fixed; test withheld operation and typed composition	"Activated" or "predictive" $\Rightarrow$ the component natively performs the operation
Transport	Attention/message patterns, causal edge effects, routing gates, donor–recipient alignment	Move or swap a natural donor while holding abstract value and downstream transform fixed; test delivery and use	High attention, edge weight, or route frequency $\Rightarrow$ transported content or causal contribution
Memory	Pulse–chase decodability, persistence curves, state/cache occupancy, reader access	Write, erase, delay, and rescue the nominated physical memory state while holding generators and later demand fixed	Persistent decodability $\Rightarrow$ native storage, necessity, or addressability
CTM coordination	Cross-role timing, coalition effects, cache/recompute and route/transform coupling	Factorial edits and ordered coalitions; compare direct with composed protocols on structural holdout	Individually causal parts $\Rightarrow$ a compositional MesoMachine

Evidence profiles, not a scalar ladder. Interpretability targets are neither a total order nor a mathematically defined lattice. We report a profile

$$\mathbf{e} = (e_{\text{read}}, e_{\text{dec}}, e_{\text{loc}}, e_{\text{use}}, e_{\text{nec}}, e_{\text{suf}}, e_{\text{act}}, e_{\text{trans}}, e_{\text{comp}}, e_{\text{id}})$$

whose coordinates are separately defined estimands with uncertainty. A componentwise partial order is used only when two profiles share frame, scale, and tolerance. No "license gap" is formed by subtracting incomparable coordinates. Readability, actionability, reconstruction, and transport can be incomparable.

Four sparsities. The evidence synthesis motivates four distinct sparsities:

$$S_{\text{act}}, \quad S_{\text{var}}, \quad S_{\text{jac}}, \quad S_{\text{path}},$$

for activations, realized causal variables, Jacobian/computation, and physical pathways. Their units and failure modes differ:

S_{act} **— activation sparsity.**
Few coordinates are nonzero in a chosen activation basis. It licenses an economy of description in that basis, not a claim about causal variables.

S_{var} **— realized-variable sparsity.**
A small set of interventionally distinguished variables accounts for the tested counterfactuals. It depends on the intervention/readout frame and does not imply a sparse physical path.

S_{jac} **— computational sparsity.**
Few input-state directions exert locally non-negligible effects on declared downstream variables, for a stated Jacobian or finite-difference scale. It can be dense even when activations are sparse.

S_{path} **— pathway sparsity.**

A small physical coalition of edges, heads, neurons, or modules carries the tested causal effect under an interaction-aware estimand. Redundancy and self-repair can make a sparse native protocol look dense under isolated ablation.

None implies another without additional assumptions. Sparse autoencoders are therefore candidate coordinate systems and sensors, not privileged MesoMachines (**??????**). The distinction is experimentally productive: a method should be evaluated against the sparsity it actually optimizes, not rewarded for a different sparsity by verbal transfer.

18.2 Multi-axis evidence coding

18.3 Multi-axis evidence coding

Every claim in §23 is coded in `supplement/evidence_atlas.csv` on axes that can *veto* it rather than merely support it: `instrument_free`, `control_saturated`, `held_out`, `replication_breadth`, and `prior_art_in_card`. The last exists because a result that restates the object's own documentation is a verification, not a finding, and a column that forces one to say so is worth the cost of maintaining it.

19 Five things the first draft assumed

An audit of this paper against its own standard found five loads being carried by statements that were never made. Four are now measured. The fifth is not measurable and is instead named, because using it silently is the error.

19.1 G0 — interpersonal comparability, the assumption under §12

Every aggregate in §12 adds one person's numbers to another's. That operation is meaningless unless the numbers live on a common scale, and **nothing in the elicitation supplies one**. Person 1 writing +8 and person 2 writing +8 are two marks on two private rulers.

What the paper actually did. It min–max normalised each person to $[0, 1]$ (§7), so that everyone's best option is worth 1 and worst worth 0. That is a *stipulation of equal stakes*: it asserts that every participant cares equally much about the whole question.

Why the stipulation is not innocent. Consider two participants: one for whom all four responses are nearly equivalent, and one for whom the choice is grave. Min–max gives them identical influence over the aggregate. A different and equally defensible stipulation — normalise by a common scale factor, so that intensity counts — gives the second participant more. The two produce different winners.

Proposition 106 (The deficiency figure is scale-relative)**.** *The Blackwell deficiency of* 0.1298 *reported in §12 is a number about the pair (data, interpersonal scale). It is not identified by the data alone, because the data contain no information distinguishing "this person's* +8 *means what that person's* +8 *means" from any other calibration.*

Why this is the deepest limitation in the paper. Arrow's theorem is precisely a theorem about what can be done *without* interpersonal comparison; Harsanyi's utilitarian representation requires it. So the choice made here silently determines which side of a hundred-year-old divide the analysis sits on. It is now stated. It cannot be resolved at this site: resolving it needs an elicitation that measures intensity on a shared anchor, which is a design requirement and appears in §28.

> **Boundary.** Every aggregate quantity in this paper — the deficiency, the rule comparisons, the maximin figures — is conditional on min–max interpersonal normalisation. The *ordinal* results (Condorcet structure, IIA rate, agreement rates) are not, because they never add across people.

19.2 G1 — ties, asserted and now measured

The paper's decision rules use $\arg\max$, and exact ties would make them ill-defined. The first draft asserted that ties occur with probability zero. Measured over all 968 prompts:

exact ties between the top two scores	**0** of 968
relative gaps below 10^{-6}	0 of 968

Why it needed checking rather than asserting. "Probability zero" is a statement about a continuous model; the data are finite and produced by a tie-prone construction (sums of a few dozen products of bounded numbers). The claim happened to be true. It was not entitled to be assumed, and an assertion that turns out true is still an assertion until it is run.

19.3 G2 — manipulability, and what it does to "sincere"

Every number in this paper reads the elicited weights as what people believe. Gibbard–Satterthwaite guarantees that any non-dictatorial rule over three or more alternatives *can* be manipulated; it says nothing about how often, and the rate is what decides whether the sincerity reading is safe.

Protocol 107 (Crudest manipulation)**.** For each (annotator, prompt): compute the collective winner honestly, then replace *all* of that annotator's weights by +10, and separately by −10, and recompute. Score both outcomes by that annotator's own stated ranking.

(annotator, prompt) pairs tested	15,482
pairs where saturating the weights yields a *strictly better* outcome	
by that annotator's own ranking	604 = **3.9**%

Remark 108 (This is a lower bound, and that is the point). The manipulation tested is the least intelligent one available — push everything to one end. A participant choosing which criteria to inflate can only do better. So 3.9% is a floor on the rate at which a participant had something to gain by misreporting, and the sincerity reading of the weights is safe only to that degree. No claim in this paper depends on sincerity being exact; several would change in magnitude if the true manipulation rate were much larger, and that is recorded in §27.

19.4 G3 — the judge scores criteria against the responses they were written about

This is a circularity the first draft did not state. The criteria were written *after* the participant saw the four responses, typically to explain why one of them is best. The judge then scores those criteria *against those same four responses*.

criterion–author pairs	95,553
satisfaction of the author's own top-ranked response	0.5831
mean satisfaction of the other three	0.5466
difference	+0.0365 (se 0.0008, z = +46.2)

How to read this, and how not to. It is not a defect of the judge. It is what "I wrote down why X is best" means: a criterion authored by someone who ranked X first is, by construction, more satisfied by X. What it establishes is that satisfaction and choice are **two views of one act**, not two independent measurements of a latent value. Every correlation between the criteria route and the ranking route — including the 0.393 of table 8 and the 68.4% of section 12.2 — is inflated by this mechanism, and the inflation cannot be removed without response-blind criteria.

19.5 G4 — the two elicitations disagree beyond their own unreliability

The criteria route and the ranking route select different winners on 31.6% of prompts (section 12.2). A natural deflation is: both are noisy, and noise alone explains it. Theorem 63 answers this exactly, because it says a correlation measured through unreliable instruments is attenuated by $\sqrt{\mathrm{rel}_1 \mathrm{rel}_2}$ — so dividing by that factor removes the noise and leaves the disagreement.

	split-half	Spearman–Brown
reliability of the ranking route	0.7702	0.8702
reliability of the criteria route	0.8882	0.9408
observed cosine between the two routes		0.6265
disattenuated = $0.6265/\sqrt{0.8702 \times 0.9408}$		**0.6924**

Theory claim. Correcting *both* routes for their own unreliability raises the agreement only from 0.626 to 0.692. The two elicitations still do not coincide. So roughly 31% of the gap between what people *write* and what they *choose* is real disagreement, not measurement error — and by G3 this is an *under*statement, because the circularity inflates the observed agreement in the first place.

Corollary 109 (The paper's largest quantity, restated at the panel level)**.** *table 8 reported* 0.393 *for one person against their own ranking. section 19.5 reports* 0.692 *for the whole panel's criteria against the whole panel's rankings, disattenuated. The two are consistent: averaging over people removes individual noise (theorem 64) and raises the number, but the residual gap does not close. Whatever the pipeline does downstream, the largest single discrepancy in the chain sits* inside *the elicitation, between two things the same people said in the same session.*

20 The estimators

20.1 Why the query family must be declared before anything is measured

Definition 110 (Query-relative equivalence)**.** Given a family Q of queries, $N \sim_Q N'$ iff $q(N, x) = q(N', x)$ for all $q \in Q$ and all inputs x. *Normative information* is the equivalence class $[N]_Q$, not the text.

Proposition 111 (Without Q, "loss" is not a defined quantity)**.** *There exist families Q under which every change is total loss, and families under which no change is any loss. Hence the answer to "how much was lost" is determined entirely by Q.*

Proof. Take $Q_{\max}$ = all measurable functions of N. Then $N \sim_Q N'$ iff $N = N'$ (the identity function is in $Q_{\max}$), so any change destroys the equivalence entirely. Take $Q_{\min} = \{\text{a constant map}\}$. Then $N \sim_Q N'$ for all pairs, so no change is a loss. Both families are admissible in the absence of a stipulation, and they give opposite answers. □

Corollary 112. *A framework that hides Q inside a supremum has* relocated *the choice, not made it. Publishing Q before measuring is not a preliminary; it is most of the result.*

20.2 The decision-theoretically correct quantity: normalised regret

Definition 113 (Normalised regret)**.** For a perturbation taking norm N to N', with $a^\star := \arg\max_a Y_N(a)$ and $a' := \arg\max_a Y_{N'}(a)$,

$$\mathrm{Reg}(N \to N') \;:=\; \frac{Y_N(a^\star) - Y_N(a')}{Y_N(a^\star) - \min_a Y_N(a)} \;\in\; [0, 1].$$

Why this and not something else. Reg is exactly the Blackwell deficiency of theorem 84 restricted to the decision family "choose one of these four, scored by the original norm's own utility". It is not a proxy for the quantity of interest; on that family it *is* the quantity. The denominator makes it invariant to the arbitrary scale of Y, which remark 73 showed is not identified.

Definition 114 (Flip rate)**.** $\pi := \mathbb{P}(a' \neq a^\star)$, estimated by the average of $\mathbf{1}\{a' \neq a^\star\}$ per equation (1) and corollary 38.

Lemma 115 (The flip rate is a coarsening of regret)**.** $\mathbf{1}\{\mathit{flip}\} = \mathbf{1}\{\mathrm{Reg} > 0\}$*, and therefore, exactly,*

$$\mathbb{E}[\mathrm{Reg}] \;=\; \pi \cdot \mathbb{E}[\mathrm{Reg} \mid \mathit{flip}].$$

Proof. If $a' = a^\star$ then the numerator of definition 113 is zero, so $\mathrm{Reg} = 0$; if $a' \neq a^\star$ then $Y_N(a') < Y_N(a^\star)$ by definition of the arg max (assuming no exact tie), so $\mathrm{Reg} > 0$. Hence the indicators coincide. Then $\mathbb{E}[\mathrm{Reg}] = \mathbb{E}[\mathrm{Reg} \mid \mathrm{Reg} > 0]\mathbb{P}(\mathrm{Reg} > 0) + 0$, by the law of total expectation, which for a finite space is just splitting the defining sum into the two events. □

operator	π	$\mathbb{E}$[Reg]	$\mathbb{E}$[Reg | flip]
double the target weight	4.2%	0.0043	0.1029
delete the target criterion	5.1%	0.0067	0.1307
invert the target weight	11.6%	0.0294	**0.2531**
halve the target weight	2.3%	0.0012	0.0532

Table 15: 968 prompts × 20 target draws. Target criterion sampled, not fixed; see section 20.4.

Corollary 116 (The discarded factor carries information)**.** *By flip rate, invert exceeds delete by* 11.6/5.1 = 2.3×*. By expected regret, by* 0.0294/0.0067 = 4.4×*. Reporting only the flip rate halves the measured separation between operators, because inversion not only flips more often, it flips* worse *–* 25.3% *of the available utility range versus* 13.1%*.*

20.3 The flip rate is capped by the margin distribution

Definition 117 (Margin)**.** $m := Y(a_{(1)}) - Y(a_{(2)})$, the gap between the top two responses.

Proposition 118 (Margin cap)**.** *Let $\delta := Y_{N'} - Y_N$ be the perturbation of the score vector. Then $\pi \le \mathbb{P}(m < 2\|\delta\|_\infty)$.*

Proof. A flip requires the perturbed score of some $a \neq a^\star$ to exceed that of $a^\star$, i.e. $Y(a) + \delta(a) > Y(a^\star) + \delta(a^\star)$, i.e. $\delta(a) - \delta(a^\star) > Y(a^\star) - Y(a) \ge m$. Since $|\delta(a) - \delta(a^\star)| \le 2\|\delta\|_\infty$, a flip implies $m < 2\|\delta\|_\infty$. Taking probabilities and applying monotonicity of $\mathbb{P}$ over the implication gives the bound. □

Measured relative margins $m/(\max_a Y - \min_a Y)$:

$$p_{05} = 0.026, \quad p_{25} = 0.144, \quad \text{median} = 0.319, \quad p_{75} = 0.527, \quad p_{95} = 0.800.$$

Corollary 119. *An operator whose perturbation is smaller than about a third of the score range cannot flip half the prompts,* however much normative content it destroys. *The ceiling on a flip rate is set by the margin distribution, not by the norm. This is the formal reason definition 113 is the primary quantity and π the secondary one.*

20.4 Target selection is an axis, not a nuisance

Every criterion-level operator must choose *which* criterion to act on. Measured effect of that choice on the induced change in Y:

$$\text{highest } |w| : 0.273 \qquad \text{random} : 0.176 \qquad \text{lowest } |w| : 0.041$$

– a ratio of 6.6×, larger than the differences between operators. Reporting a single cell would therefore report the selection rule. All numbers in this paper sweep four selection rules (highest $|w|$, lowest $|w|$, most-rated, random) × five seeds, and the seeds are verified to change the draws: only 0.1% of targets are shared across all five.

20.5 Variance of these estimators

With $P = 968$ prompts and $R = 20$ draws each, theorem 57 applies with measured $\sigma_b^2 = 0.01361$, $\sigma_w^2 = 0.03677$:

$$\rho = 0.2701, \qquad \text{DEFF} = 1 + 19(0.2701) = 6.13, \qquad \sqrt{\text{DEFF}} = 2.48.$$

Remark 120 (Derivation checked against an independent measurement). A resampling estimate of the same inflation, obtained by recomputing every pairwise statistic at the cluster level and comparing to the pooled version, gave 2.79. Derived 2.48, measured 2.79: the residual is that the pairwise statistic is a correlation rather than a simple mean, for which theorem 57 is only approximate.

operator	π	se	95% CI	MDE
double	4.2%	0.0036	[3.5, 4.9]%	1.00pp
delete	5.1%	0.0037	[4.4, 5.8]%	1.05pp
invert	11.6%	0.0052	[10.6, 12.6]%	1.46pp
halve	2.3%	0.0026	[1.8, 2.8]%	0.72pp
paired delete − double	+0.93pp	0.0028	$z = +3.3$	0.79pp

Table 16: Standard errors from the $P = 968$ cluster means, per equation (2). MDE from theorem 54.

Corollary 121 (Sample size)**.** *By corollary 55 with $\sigma_b = \sqrt{0.01361 + 0.03677/20} = 0.1243$, resolving a 1pp difference at 80% power requires $P = (2.8016 \times 0.1243/0.01)^2 = 1{,}213$ prompts. The release has 968: the design is about 20% short of its own headline resolution, and the delete-versus-double difference clears its MDE only barely.*

21 The causal layer: influence is not agreement

21.1 Why a correlational measure cannot answer this

Definition 122 (Preservation, causal form)**.** A distinction reached the output iff *removing it changes the output*.

A participant whose view happens to match the majority scores perfect alignment while contributing nothing: remove them and the outcome is unchanged. A participant who moved the outcome may sit far from it. Alignment and influence are therefore different functionals, and only the second answers the definition.

21.2 The Shapley value, derived

Definition 123 (Characteristic function)**.** For prompt p with annotator set A and winner $w^\star := \arg\max_r \sum_{a\in A} N_a(r)$, define for $S \subseteq A$

$$v(S) \;:=\; \mathbf{1}\Big\{\arg\max_r \textstyle\sum_{a\in S} N_a(r) = w^\star\Big\}.$$

Definition 124 (Shapley value)**.**

$$\phi_i \;:=\; \sum_{S\subseteq A\setminus\{i\}} \frac{|S|!\,(|A|-|S|-1)!}{|A|!}\,\big[v(S\cup\{i\}) - v(S)\big].$$

What the weights mean. Consider all $|A|!$ orderings of the annotators. The bracket is i's marginal contribution when it arrives after exactly the set S. The number of orderings in which that happens is $|S|!\,(|A|-|S|-1)!$ – $|S|!$ ways to arrange those before, $(|A|-|S|-1)!$ ways to arrange those after. Dividing by $|A|!$ makes the weights a probability distribution over orderings. So ϕ_i is **the average marginal contribution of i over a uniformly random arrival order**, which is exactly the causal definition above, averaged over every context in which i could arrive.

Estimation. The sum has $2^{|A|-1}$ terms. We sample 200 random orderings per prompt and average the realised marginal contributions, which is an unbiased estimator of ϕ_i by construction (each ordering is drawn from the same uniform distribution the weights encode).

21.3 Result: the pre-registered kill does not fire

1,865 (annotator, prompt) pairs	
Corr(Shapley influence, cosine alignment)	+0.3040
bootstrap 95% CI (protocol 58)	[+0.2566, +0.3520]
pre-registered kill: correlation ≈ 1.0	**does not fire**

alignment quintile	mean alignment	mean influence
Q1	−0.0416	0.0083
Q2	+0.8164	0.1055
Q3	+0.9489	**0.1325**
Q4	+0.9861	0.1262
Q5	+0.9973	0.1262

Table 17: Influence rises steeply from Q1 to Q3 and then *flattens*.

Theory claim. Beyond moderate agreement, agreeing more buys no influence at all. And the off-diagonal cells are populated: 0.75% of pairs are top-quintile aligned with bottom-quintile influence, 3.11% are bottom-aligned with top-quintile influence. "Agreement is not preservation" is here a count of real people, not a slogan.

21.4 An independent confirmation from a different operator

Dropping the *most conforming* annotator changes the decision on 5.8% $[4.3, 7.1]$ of prompts; dropping the *largest dissenter*, on 2.1% $[1.1, 3.2]$. The intervals do not overlap and the sign is stable across all three judges. The dissenter was already not influencing the outcome – the same statement as the flat top of table 17, reached by a different route.

22 What is not identified, with proofs

Proposition 125 (Contamination and redundancy are not separable here)**.** *Let α_p be the alignment between a prompt's weighted-criteria score and the crowd's ranking, and s_p the sensitivity of its decision to perturbing one criterion. Two models,*

C: criteria written after *seeing responses are individually more decisive, so a high-α rubric absorbs single-criterion perturbations;*

R: a high-α rubric is one whose criteria agree with each other, and a redundant rubric is insensitive to any single member,

imply the same sign for $\mathrm{Corr}(\alpha, s)$*, and the observed data distribution is a function of* (α, s) *alone. The likelihoods coincide; the models are not identified.*

What is identified: a bound. Splitting prompts at the median of α:

operator	low α	high α	ratio
double	5.1%	2.8%	0.55×
delete	5.6%	4.6%	0.82×
invert	13.1%	9.9%	0.75×

Corollary 126. *Every flip rate reported here varies by up to* 1.8× *across the range of* α*. That bound belongs on every such number. Separating C from R requires response-blind elicitation and cannot be done at this site.*

Proposition 127 (The probe bound is an identification limit)**.** *By theorem 74, at most two parameters per person per prompt are identified from the ratings. Everything else in the criterion text is unidentified from the released quantitative data.*

Proposition 128 (Compilation's three actions are confounded)**.** *Selection, rewriting and weight-discarding occur in one step with no lineage. Their individual contributions to the* 29.4% *disagreement are not identified.*

Proposition 129 (Y does not exist)**.** *The chain in equation (4) terminates at* D. Y *– what a system trained under this standard does – is not in the release, and no arithmetic recovers it. Any end-to-end number quoted to* Y *is fabricated.*

23 Findings

Each finding ends with its confidence card. A missing entry reads UNCOMPUTED; none is omitted.

23.1 F1 — the elicitation captured consensus, not the individual

$\cos(N_i, \text{own ranking}) = 0.3928$ versus $\cos(N_i, \text{stranger's ranking}) = 0.3097$; difference +0.0831.

n_{eff}	968 prompt clusters (14,764 person–prompt rows)
MDE	0.0187 in cosine, from theorem 54 at the observed se
effect / floor	0.0831/0.0067 = 12.4
CI width / |eff|	0.0262/0.0831 = 0.32
spec_survival	UNCOMPUTED (single specification)
seed spread / |eff|	not applicable (no stochastic step)
held out	ABSENT
instrument	routes through the judge; ceiling 0.720 (§15)
prior art	none located; the release documents no such comparison
multiplicity	1 test in this family

23.2 F2 — no finite weight encodes a veto

theorem 90 and theorem 91; required M up to 26.54 against a scale maximum contribution of 10.

n_{eff}	161 prompts with a partial veto; 42 binding
MDE	not applicable — this is a **derivation**, not a test
effect / floor	not applicable
CI width / |eff|	not applicable
spec_survival	the theorem holds for every score construction
seed spread	not applicable (deterministic)
held out	the *maximum* is what carries it; a held-out set can only raise it
instrument	Δ is computed from the judge tensor; the *theorem* is not
prior art	the incompatibility of vetoes with scoring rules is classical; the *measured* Δ distribution on this release is not
multiplicity	none — a single derived statement

Remark 130 (Labelled as a derivation, per the arithmetic trap). theorem 91 could not have come out otherwise: it follows from theorem 90 plus the unboundedness of Δ. It is stated as a theorem, its assumption is named (S can approach its bounds independently across responses), and it is not counted as evidence in the sense a measurement is.

23.3 F3 — comparative measurements are instrument-invariant; absolute ones are not

table 11, table 12.

n_{eff}	968 prompts (three instruments); 300 for the reordering
MDE	ρ resolution ≈ 0.15 at 11 operators
effect / floor	orderings 0.963–1.000 against a chance ρ of 0
CI width / |eff|	bootstrap intervals in table 16
spec_survival	3 of 3 instruments preserve the ordering
seed spread / |eff|	< 0.1 across 5 seeds of the target draw
held out	the reordered run is a genuine held-out instrument
instrument	this finding *is* about the instrument, by construction
prior art	judge-sensitivity of LLM evaluation is known; the comparative/absolute split measured on this object is not
multiplicity	3 instruments $\times$ 11 operators; the reported statistic is the ordering, one per instrument

23.4 F4 — influence and alignment are distinct (C1 survives)

Corr = +0.3040 [+0.2566, +0.3520], with the quintile profile of table 17.

n_{eff}	968 prompt clusters (1,865 annotator–prompt rows)
MDE	0.065 in correlation, at the bootstrap se
effect / floor	$0.304/0.024 = 12.7$
CI width / |eff|	$0.095/0.304 = 0.31$
spec_survival	UNCOMPUTED; one characteristic function tested
seed spread / |eff|	< 0.02 over the 200-permutation Shapley estimator
held out	ABSENT
instrument	routes through the judge; ceiling 0.720
prior art	Shapley-versus-alignment has not been reported on this object
multiplicity	1 pre-registered test

23.5 F5 — one bit per criterion reproduces 89.8% of decisions

section 14.1, cross-checked against table 7.

n_{eff}	968 prompts
MDE	± 1.0pp at the observed se
effect / floor	the quantisation is deterministic; no resampling floor applies
CI width / |eff|	UNCOMPUTED
spec_survival	agrees with the independent sign-only test to 1.5pp
seed spread	not applicable (deterministic)
held out	ABSENT
instrument	judge-dependent in level; the *bit curve* is not tested across judges
prior art	none located for this object
multiplicity	5 quantisation levels, reported in full

24 What did not survive

A paper with no retractions is a paper whose ontology has no word for error. All entries are cross-referenced to `supplement/exclusion_log.csv`.

withdrawn claim	what killed it	status
"the operator design has rank 2, so all remaining dimensions require the judge"	the rank statistic sits inside the span of three defensible nulls (12.1, 17.0, 18.0) with the observation at 13; the null *construction* moved it more than the data	rank retired as a statistic
"strengthening a criterion moves the decision, removing one does not"	direct measurement: deleting flips 8.8%, doubling 4.2%. The asymmetry was an artefact of normalising the score *after* mutating and then differencing two separately normalised vectors	withdrawn, sign and all
"minority-sensitive aggregation changes half the outcomes"	the sham control (same people, matched moments, response pattern destroyed) disagrees 61.4% versus the real 49.4%	withdrawn; the residual runs the other way
"aggregation is a 10× larger lever than the criterion layer"	the aggregators were indexed over criteria rather than people	magnitude withdrawn; ordering survives at 4.3×
"the flip rate is confounded by decision margin"	measured Q_1/Q_4 ratios of 1.0, 1.0, 1.1	defect opened by us and closed by its own measurement
"wide margins arise because agreeing criteria carry larger weights"	$\mathrm{Corr}(\text{margin}, \overline{\lvert w\rvert}) = +0.002\ [-0.067, +0.059]$	mechanism refuted; the flatness stands unexplained

Remark 131 (The failure mode behind most of these). Six of the six are the same shape: a verdict decided by a constant the author chose – one null instead of a range, one reference instead of a class, a two-branch template for a three-outcome test, a 0.15 cutoff. The repair adopted here is structural: *every verdict is a printed comparison against the range of defensible references, never a boolean computed from a threshold.*

25 Falsification and contradiction handling

Stated before the claims are defended.

1. **F1 dies** if a response-blind elicitation shows $\cos(N_i, \text{own})$ far exceeding $\cos(N_i, \text{stranger})$. The present gap would then be an artefact of post-hoc writing rather than a property of elicitation.
2. **F2 dies** if a bounded M is exhibited that reproduces constraint behaviour across candidate sets. theorem 91 forbids it for unbounded S; a bounded-satisfaction variant with a stated bound would restrict the theorem's scope.
3. **F3 dies** if a fourth instrument reverses the operator ordering ($\rho < 0.8$). Three have not.
4. **F4 dies** if $\mathrm{Corr}(\text{influence}, \text{alignment}) \to 1$ under a different characteristic function. Only one has been tested; this is the weakest of the five.
5. **F5 dies** if the bit curve differs materially under another judge. Not tested.
6. **The framing itself dies** if a person's induced ordering is unstable across response sets. Then what was elicited is a property of the probe, not of the person, and nothing downstream measures values at all. This is the deepest available test and it requires two candidate sets per person, which this release does not have. corollary 81 (13.4% IIA violations) is a lower bound on the concern, not a resolution of it.

26 Discussion

26.1 The claim, and its novelty boundary

What is not new. Blackwell's theorem, Arrow's impossibility, Sen's liberal paradox, the Shapley value, Spearman–Brown, the design effect, attenuation from measurement error. The incompatibility between vetoes and scoring rules is classical social choice. Judge-sensitivity of language-model evaluation is documented. None of these is claimed here.

What is new, and only this. (i) The measured Δ distribution that turns the veto incompatibility into a *quantitative* statement about a specific release, with the counterexample present in the data (section 13.2). (ii) The self-versus-stranger decomposition showing that the individual signal in this elicitation is 0.083 of cosine (table 8). (iii) The comparative/absolute split under the gauge test (§15.3). (iv) The bit-rate of the weight scale (section 14.1). (v) The demonstration that the weighted-criteria score agrees with the Condorcet winner only 68.4% of the time while every ranking rule agrees 97–98%.

26.2 Why method relativity is not relativism

Every quantity here is relative to a declared query family (proposition 111), a declared decision family (definition 113), and a named instrument (§15). That is not an admission that the numbers are arbitrary; it is the statement that they are *typed*. A claim inherits the scope of the family it was computed against and may not be quoted outside it. The gauge test (§15.3) is what makes the typing empirical rather than rhetorical: it shows exactly which claims move when the instrument moves.

27 Limitations and epistemic status

"We did not run X" and "X cannot be run at this site" are different sentences and are not merged.

27.1 Structural — cannot be run here

1. **Response-blind source norms.** Every criterion was written after seeing the responses. proposition 125 shows contamination and redundancy are not separable without them.
2. **A second candidate set per person.** Required to test theorem 91 out of sample and to resolve the probe-dependence question of §25(6).
3. **Typed fields.** 82.3% of criteria carry none (table 9); scope and exception loss are unstudiable here.
4. **Compilation lineage.** Absent, so §14's three actions stay confounded.
5. Y**.** Not in the release.
6. **Cross-dataset / cross-domain.** One release exists.

27.2 Practical — could be run, was not

1. Held-out partitions for F1, F4, F5.
2. A specification curve over characteristic functions for the Shapley estimator; one was tested.
3. The bit curve of section 14.1 across judges.
4. Two-way clustering (theorem 65) on the person-level quantities; only prompt-level clustering was applied, and annotators repeat across prompts.

27.3 Quantitative bounds that apply to every number here

bound	value
judge attenuation ceiling	0.741
panel reliability ceiling	0.9722
compounded ceiling on any correlation	0.720
contamination-proxy variation in every flip rate	up to 1.8×
absolute winner claims: instrument share	≈ 40%
absolute winner claims: presentation-order share	8.0pp
rubric corpus never joined	1.8%
design shortfall for 1pp resolution	968 of 1,213 prompts
pairwise-dependence screen: MDE at full n	$\lvert\phi\rvert = 0.153$ against median 0.124

Remark 132 (The last row read honestly). The median pairwise association in the operator grid is *below* the MDE at the largest available sample. The pairwise table is therefore a screen for *strong* dependence and is worded that way throughout; its nulls are underpowered rather than informative.

28 What a design that can measure this requires

Each row is derived from a specific result above, not proposed.

requirement	derived from	unlocks
response-blind source norms	proposition 125	the only route separating contamination from redundancy
$\geq$ 2 candidate sets per person	theorem 91, corollary 81	tests whether M transfers; quantifies candidate-set overfitting
typed fields: veto, scope, exception, priority as separate columns	§13, table 9	the index mismatch is a missing column, not a hard problem
interval-scale validation, or paired comparisons instead	table 7	11.7% of decisions currently rest on an unestablished assumption
$k > 4$ responses	theorem 74	raises the probe bound from 2 to $k - 2$
$\geq$ 1,213 prompts	corollary 55	1pp resolution
substantially more than 968 for pairwise structure	§27	the typical association is currently undetectable
$\geq$ 3 executors	§15.3	already available; the gauge is established
compilation receipt (lineage)	§14	separates selection, rewriting, weight-discarding
declared Q and declared A	proposition 111, remark 68	without them "loss" is undefined and "defect" is inseparable from "social choice"

29 Conclusion

The chain of equation (4) loses something at every arrow, and the losses are not of one kind.

- $\succeq_i \to N_i$**.** An unbounded relation is projected onto two free parameters (theorem 74) – forced, not recoverable. Within that, a person's own criteria recover 39% of the direction of their own choice, and the individual component of that is 0.083 of cosine. The type structure is empty: 82.3% of criteria carry no condition, exception, priority, hedge or prohibition. And 11.7% of decisions depend on an interval reading of a scale that was never validated as one.
- $N_i \to G$**.** Aggregation costs 0.1298 of normalised utility (theorem 84) – a social choice, but one made silently because A was never declared. The weighted-criteria route agrees with the Condorcet winner only 68.4% of the time while ranking rules agree 97–98%. 13.4% of pairwise relations flip when the option set changes. And 26.1% of the time the utilitarian winner is a response somebody called unacceptable.
- **Type.** No finite weight encodes a veto (theorem 91). This is the strongest statement in the paper and it required no instrument.
- $G \to C$**.** Compilation changes 29.4% of decisions, with its three actions confounded. But the discarded magnitudes were worth about one bit: sign alone reproduces 89.8%.
- $C \to D$**.** The executor's reliability is 0.5497, capping every correlation at 0.741. Changing the reader changes 40% of winners; reordering the page changes 8%. Comparative claims survive all of it; absolute claims do not.
- $D \to Y$**.** Does not exist in the release.

Theory claim. The losses inside the elicitation are larger than every loss the pipeline adds. The most defensible statement this object supports is a *theorem* about what a scalar scale cannot encode, and the most consequential one is that the next dataset should buy **fields and response-blindness**, not finer weights or more prompts.

References